

Online Supplement for
‘Modal Estimation and Inference of Autoregressive Distributed Lag Model
Applied to COVID-19 Pandemic Data’

by

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This Online Supplement is an Appendix that supplements the claims, simulations and empirical analysis provided in the main text.

A Appendix

This supplementary Appendix comprises five sections. Section A.1 derives the moments of the modal score, and Section A.2 provides auxiliary lemmas to facilitate the proofs of the main claims. Section A.3 proves the main claims, and Sections A.4 and A.5 provide supplementary simulations and empirical information.

A.1 Moments of the Modal Score

In this section, we derive the moments of the modal score. Throughout, we let $\mu_2 := \int t^2 \phi(t) dt$, which equals unity for the standard normal kernel.

We first derive the approximation of m_h . By the change of variables, if we let $u = hs$,

$$m_h = \mathbb{E}[\phi^{(1)}(u_{th})] = \int \phi^{(1)}(u/h) f_u(u) du = h \int \phi^{(1)}(s) f_u(hs) ds.$$

Assumption 1 (iii.a) permits the following expansion:

$$f_u(hs) = f_u(0) + hf_u^{(1)}(0) + \frac{1}{2}h^2s^2f_u^{(2)}(0) + \frac{1}{6}h^3s^3f_u^{(3)}(0) + o(h^3s^3), \quad (\text{A.1})$$

and for $j = 0, 1, 2, 3$ and $\ell = 1, 2, 3$, $\int |s|^j |\phi^{(\ell)}(s)| ds < \infty$ by Assumption 1 (ii.a) if $j \leq 2$, which leads to $|\phi^{(\ell)}(s)| = o(|s|^{-4-\lambda})$ as $|s| \rightarrow \infty$, so that for $j = 0, 1, 2, 3$, $|s|^j |\phi^{(\ell)}(s)| = o(|s|^{-1-\lambda})$. We obtain

$$\int \phi^{(1)}(s) ds = 0, \quad \int s \phi^{(1)}(s) ds = -1, \quad \int s^2 \phi^{(1)}(s) ds = 0, \quad \text{and} \quad \int s^3 \phi^{(1)}(s) ds = -3\mu_2$$

by integration by parts, where the first and third integrals vanish because $\phi^{(1)}(\cdot)$ and $(\cdot)^2\phi^{(1)}(\cdot)$ are odd functions. For the second and fourth integrals, integration by parts gives rise to $\int s^m\phi^{(1)}(s)ds = [s^m\phi(s)]_{-\infty}^{\infty} - m \int s^{m-1}\phi(s)ds = -m \int s^{m-1}\phi(s)ds$, where the last equality follows from Assumption 1 (ii.a). If we let $m = 1$ and 3 , the second and fourth integrals follow, respectively. As $f_u^{(1)}(0) = 0$ by the first-order condition, the contributions driven by $O(h)$ and $O(h^2)$ drop out, so that it follows that

$$m_h = h \left\{ -\frac{1}{2}h^3\mu_2f_u^{(3)}(0) + o(h^3) \right\} = -\frac{h^4}{2}\mu_2f_u^{(3)}(0) + o(h^4). \quad (\text{A.2})$$

Second, we derive the approximation of $\text{var}(\bar{\phi}^{(1)}(u_{th})/\sqrt{h})$. By the same change of variables, we note that

$$\mathbb{E}[\{\phi^{(1)}(u_{th})\}^2] = \int [\phi^{(1)}(u/h)]^2 f_u(u) du = h \int [\phi^{(1)}(s)]^2 f_u(hs) ds.$$

Therefore, if we combine this with (A.1), then

$$\begin{aligned} & \int [\phi^{(1)}(s)]^2 f_u(hs) ds \\ &= \int [\phi^{(1)}(s)]^2 \left[f_u(0) + hs f_u^{(1)}(0) + \frac{1}{2}h^2 s^2 f_u^{(2)}(0) + \frac{1}{6}h^3 s^3 f_u^{(3)}(0) + o(h^3 s^3) \right] ds \end{aligned}$$

Here, $\int [\phi^{(1)}(s)]^2 s ds = 0$ and $\int [\phi^{(1)}(s)]^2 s^3 ds = 0$, because $[\phi^{(1)}(\cdot)]^2$ is even from the fact that $\phi^{(1)}(\cdot)$ is odd. Furthermore, $v^2 := \int [\phi^{(1)}(s)]^2 ds < \infty$ by Assumption 1 (ii.b) and $\int s^2 [\phi^{(1)}(s)]^2 ds < \infty$ by Assumption 1 (ii.a), so that it follows that

$$\mathbb{E}[\{\phi^{(1)}(u_{th})\}^2] = h \int [\phi^{(1)}(s)]^2 f_u(hs) ds = hv^2 f_u(0) \{1 + O(h^2)\}. \quad (\text{A.3})$$

Given this, $m_h^2 = O(h^8)$ from (A.2), so that

$$\text{var}\left(\frac{\bar{\phi}^{(1)}(u_{th})}{\sqrt{h}}\right) = \frac{1}{h} \left(\mathbb{E}[\{\phi^{(1)}(u_{th})\}^2] - m_h^2 \right) \rightarrow v^2 f_u(0),$$

as $h \rightarrow 0$.

Third, we derive the corresponding moments of the second- and third-order kernel derivatives, which we use for Lemmas A.5 and A.7 given below. For $j = 2, 3$, the same change of variables gives $\mathbb{E}[\phi^{(j)}(u_{th})] = h \int \phi^{(j)}(s) f_u(hs) ds$, and repeated integrations by parts yield the following:

$$\int \phi^{(2)}(s) ds = 0, \quad \int s \phi^{(2)}(s) ds = 0, \quad \int s^2 \phi^{(2)}(s) ds = 2, \quad \int s^3 \phi^{(2)}(s) ds = 0,$$

$$\int \phi^{(3)}(s)ds = 0, \quad \int s \phi^{(3)}(s)ds = 0, \quad \int s^2 \phi^{(3)}(s)ds = 0, \quad \int s^3 \phi^{(3)}(s)ds = -6. \quad (\text{A.4})$$

From the fact that $\phi(\cdot)$ is even, $\phi^{(2)}(\cdot)$ is even, but $\phi^{(1)}(\cdot)$ and $\phi^{(3)}(\cdot)$ are odd, so that $(\cdot)\phi^{(2)}(\cdot)$, $(\cdot)^3\phi^{(2)}(\cdot)$, $\phi^{(3)}(\cdot)$, and $(\cdot)^2\phi^{(3)}(\cdot)$ are odd, letting their integrals vanish. The remaining equalities, viz., $\int \phi^{(2)}(s)ds$, $\int s^2\phi^{(2)}(s)ds$, $\int s\phi^{(3)}(s)ds$, and $\int s^3\phi^{(3)}(s)ds$ follow by repeated integrations by parts. Substituting (A.1) into $h \int \phi^{(j)}(s)f_u(hs)ds$, we obtain

$$\mathbb{E}[\phi^{(2)}(u_{th})] = h^3 f_u^{(2)}(0) + o(h^4) = h^3 f_u^{(2)}(0)\{1 + o(h)\} \quad \text{and}$$

$$\mathbb{E}[\phi^{(3)}(u_{th})] = -h^4 f_u^{(3)}(0) + o(h^4) = O(h^4) \quad (\text{A.5})$$

using $f_u^{(1)}(0) = 0$. Here the relative error is not $O(h^2)$ but $o(h)$. Under Assumption 1 (iii.a), $f_u(\cdot)$ is three times continuously differentiable, so that the Taylor remainder in (A.1) is of $o(h^3 s^3)$, and the third-order term drops out because $\int s^3 \phi^{(2)}(s)ds = 0$.

Finally, the second moment of $\phi^{(j)}(u_{th})$ is obtained similarly to the second moment of $\phi^{(1)}(u_{th})$ for $j = 1, 2, 3$. That is,

$$\mathbb{E}[\{\phi^{(j)}(u_{th})\}^2] = h \int [\phi^{(j)}(s)]^2 f_u(hs) ds = h v_j^2 f_u(0)\{1 + O(h^2)\}, \quad v_j^2 := \int [\phi^{(j)}(s)]^2 ds. \quad (\text{A.6})$$

Here, $O(h^2)$ and $O(h^4)$ terms drop out as $[\phi^{(j)}(\cdot)]^2$ is even. Because $\mathbb{E}[\phi^{(2)}(u_{th})]^2 = O(h^6)$ and $\mathbb{E}[\phi^{(3)}(u_{th})]^2 = O(h^8)$, they are of smaller order than h , so that $\text{var}[\phi^{(2)}(u_{th})] = O(h)$ and $\text{var}[\phi^{(3)}(u_{th})] = O(h)$. We use these features while proving of Lemma A.5. Note that $v_1^2 = v^2$ by construction.

A.2 Auxiliary Lemmas

Before proving the main claims in the text, we first provide some preliminary lemmas and prove the main claims efficiently.

Lemma A.1. *Given Assumption 1,*

- (i) $\sum_{t=1}^n \mathbf{w}_{t-i} = O_{\mathbb{P}}(\sqrt{n})$ for $i = 0, 1, 2, \dots$;
- (ii) $\sum_{t=1}^n \bar{\mathbf{k}}_t = O_{\mathbb{P}}(\sqrt{n})$ and $\sum_{t=1}^n \mathbf{k}_t = O_{\mathbb{P}}(n)$. □

Lemma A.2. *Given Assumption 1,*

- (i) $n^{-1} \sum_{t=1}^n \mathbf{w}_{t-i} \mathbf{w}'_{t-j} \xrightarrow{\mathbb{P}} \mathbb{E}[\mathbf{w}_{t-i} \mathbf{w}'_{t-j}]$ for $i, j = 0, 1, \dots, q-1$;
- (ii) $\sum_{t=1}^n \mathbf{w}_{t-i} \mathbf{x}'_t = O_{\mathbb{P}}(n)$ and $\sum_{t=1}^n u_{t-i} \mathbf{x}_t = O_{\mathbb{P}}(n^{3/2})$ for $i = 0, 1, \dots$;
- (iii) $n^{-1} \sum_{t=1}^n \mathbf{k}_t \mathbf{k}'_t \xrightarrow{\mathbb{P}} \mathbb{E}[\mathbf{k}_t \mathbf{k}'_t]$;

(iv) $n^{-1} \sum_{t=1}^n \mathbf{k}_t \mathbf{w}'_{t-i} \xrightarrow{\mathbb{P}} \mathbb{E} [\mathbf{k}_t \mathbf{w}'_{t-i}]$ for $i = 0, 1, \dots, q-1$;

(v) $n^{-3/2} \sum_{t=1}^n \mathbf{k}_t \mathbf{x}'_t \rightsquigarrow \mathbb{E} [\mathbf{k}_t] \int_0^1 \overline{\mathbf{B}}_w'$.

□

Lemma A.3. Given Assumption 1,

(i) $n^{-3/2} \sum_{t=1}^n \mathbf{x}_t \rightsquigarrow \int_0^1 \overline{\mathbf{B}}_w$;

(ii) $n^{-2} \sum_{t=1}^n \mathbf{x}_t \mathbf{x}'_t \rightsquigarrow \int_0^1 \overline{\mathbf{B}}_w \overline{\mathbf{B}}_w'$.

□

Remark A.1. (i) Lemmas A.1 to A.3 are established by applying FCLT and the continuous mapping theorem to Assumption 1 (iv) and (vi).

(ii) Lemma A.1 describes the growth rates of the sums of the stochastic processes $\mathbf{w}_{t-i}$ and $\mathbf{k}_t$, which are $O_{\mathbb{P}}(\sqrt{n})$ and $O_{\mathbb{P}}(n)$, respectively. Lemma A.2 further explores the asymptotic behaviors between the co-products of the stochastic processes in Lemma A.1, revealing their limits. Lemma A.3 characterizes the asymptotic weak limits of the sums of $\mathbf{x}_t$ and its square. The normalization rates are $n^{3/2}$ and n^2 for $\sum \mathbf{x}_t$ and $\sum \mathbf{x}_t \mathbf{x}'_t$, respectively, reflecting the natures of integrated series $\mathbf{x}_t$, being the source of the super-consistency of the long-run parameter estimators in Theorem 2 (i) and Theorem 3. □

Let $\tilde{\mathbf{x}}'_t$ be the t -th row vector of $\widetilde{\mathbf{X}} := \mathbf{P}_{\widehat{\mathbf{W}}} \mathbf{X} = [\mathbf{I} - \widehat{\mathbf{W}}(\widehat{\mathbf{W}}'\widehat{\mathbf{W}})^{-1}\widehat{\mathbf{W}}']\mathbf{X}$.

Lemma A.4. Given Assumption 1,

(i)

$$\mathbf{D}_G^{-1} \mathbf{G}' \mathbf{G} \mathbf{D}_G^{-1} \rightsquigarrow \begin{bmatrix} 1 & \mathbf{0}' & \int_0^1 \overline{\mathbf{B}}_w' \\ \mathbf{0} & \mathbb{E} [\bar{\mathbf{w}}_t \bar{\mathbf{w}}_t'] & \mathbf{0} \\ \int_0^1 \overline{\mathbf{B}}_w & \mathbf{0} & \int_0^1 \overline{\mathbf{B}}_w \overline{\mathbf{B}}_w' \end{bmatrix},$$

where $\mathbf{D}_G := \text{diag}([\sqrt{n} \boldsymbol{\iota}'_{1+qk}, n \boldsymbol{\iota}'_k])'$, $\mathbf{G} := [\mathbf{g}_1, \dots, \mathbf{g}_n]'$, and $\mathbf{g}_t := [1, \bar{\mathbf{w}}'_t, \mathbf{x}'_t]'$;

(ii) $h^{-1/2} \mathbf{D}_G^{-1} \mathbf{G}' \overline{\boldsymbol{\Phi}}^{(1)}(\mathbf{u}_h) \rightsquigarrow [\mathcal{B}_\phi(1), \mathcal{B}_{\phi \cdot \mathbf{w}}(1)', \int \overline{\mathbf{B}}_w' d\mathcal{B}_\phi(r)]'$, where $\overline{\boldsymbol{\Phi}}^{(1)}(\mathbf{u}_h) := [\bar{\phi}^{(1)}(u_{1h}), \dots, \bar{\phi}^{(1)}(u_{nh})]'$;

(iii) $\mathbf{D}_H^{-1} \mathbf{G}' \mathbf{k} = O_{\mathbb{P}}(1)$, where $\mathbf{D}_H := \text{diag}([n \boldsymbol{\iota}'_{1+qk}, n^{3/2} \boldsymbol{\iota}'_k])'$ and $\mathbf{k} := [\mathbf{k}_1, \dots, \mathbf{k}_n]'$;

(iv) $n^{-2} \widetilde{\mathbf{X}}' \widetilde{\mathbf{X}} = n^{-2} \mathbf{X}' \mathbf{P}_{\widehat{\mathbf{W}}} \mathbf{X} = n^{-2} \mathbf{X}' [\mathbf{I} - \widehat{\mathbf{W}}(\widehat{\mathbf{W}}'\widehat{\mathbf{W}})^{-1}\widehat{\mathbf{W}}'] \mathbf{X} \rightsquigarrow \int_0^1 \widetilde{\mathbf{B}}_w \widetilde{\mathbf{B}}_w'$;

(v) $(n^2 h)^{-\frac{1}{2}} \sum_{t=1}^n \phi^{(1)}(u_{th}) \tilde{\mathbf{x}}_t \rightsquigarrow \int \widetilde{\mathbf{B}}_w d\mathcal{B}_\phi$;

(vi) $(nh)^{-\frac{1}{2}} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \mathbf{j}_t \overset{A}{\rightsquigarrow} N(\mathbf{0}, v^2 f_u(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}_t'])$.

□

Remark A.2. (i) Lemma A.4 is the MARDL counterpart of corollary A1 in Cho et al. (2015) that derives the weak limits of the normalized matrices. Lemma A.4 (ii) is the key result establishing the FCLT for the modal regression score $\phi^{(1)}(u_{th})/\sqrt{h}$ with other integrated regressors. In particular, the normalized and centered modal score $h^{-1/2} \bar{\phi}^{(1)}(u_{th})$ plays the role parallel to the quantile score

$\psi_\tau(u_t) := \tau - \mathbb{1}\{u_t < 0\}$ in the QARDL framework. Both scores are MDAs with respect to $\mathcal{F}_t$, and the limiting variance $v^2 f_u(0)$ of the former corresponds to $\tau(1 - \tau)$ of the latter. The essential difference is in the fact that the modal score is a triangular array indexed by the bandwidth, so that all weak convergence results in Lemma A.4 are array limits driven by Assumption 1 (vi).

- (ii) Lemmas A.4 (ii) and (v) provide the weak limits of the modal score along with other integrated series, and the weak limit in Lemma A.4 (v) is the source of the mixed normal limits in Theorem 2 (i) and Theorem 3. Lemmas A.4 (iv)–(vi) constitute the weak limits in Lemma A.6 given below, from which the asymptotic distributions of all parameter estimators in Theorems 1 to 5 are derived. $\square$

Lemma A.5. *Given Assumption 1,*

- (i) $\|\hat{\rho}_n - \rho_\dagger\| = O_{\mathbb{P}}((nh^3)^{-1/2});$
- (ii) $\|\hat{\kappa}_n - \beta_*\| = O_{\mathbb{P}}((n^2 h^3)^{-1/2});$
- (iii) $\|\hat{\alpha}_n - \alpha_*\| = O_{\mathbb{P}}((nh^3)^{-1/2}).$

$\square$

Remark A.3. (i) By Lemma A.5, the convergence rates of the estimators differ between the parameters

due to the mixed stationarity structure of z_t . For λ_* and ϕ_* , their estimators are influenced by two distinct sources. First, the standard deviation of the modal estimator along the stationary directions introduces an estimation error with convergence rate $(nh^3)^{-1/2}$. Second, the smoothing bias generated by the bandwidth in the modal kernel objective function introduces another estimation error with convergence rate h^2 . Under the given bandwidth condition $nh^7 = O(1)$, the latter is dominated by the former, since $h^2 = O((nh^3)^{-1/2})$, leading to the same convergence rate for the estimators of λ_* and ϕ_* . Here, $\hat{\rho}_n$ estimates $\rho_\dagger = (\nu_\dagger, v'_\dagger, \phi'_\dagger)'$ such that $\nu_\dagger = \alpha_* + \mu_* \iota'_p \phi_*$, $v_\dagger = \delta_* + \Xi_* \phi_*$, and $\phi_\dagger = \phi_*$ by (13). Therefore, $\hat{\rho}_n$ is a linear transformation of $\hat{\lambda}_n$ and $\hat{\phi}_n$ as given in (14). Consequently the convergence rate of $\hat{\rho}_n$ is determined by those of $\hat{\lambda}_n$ and $\hat{\phi}_n$.

- (ii) For the long-run parameter γ_* associated with the integrated regressor x_t , $\hat{\kappa}_n$ exhibits a rate $(n^2 h^3)^{-1/2}$ faster than that of $\hat{\rho}_n$, reflecting that $\sum_{t=1}^n x_t x'_t = O_{\mathbb{P}}(n^2)$. This is the MARDL analogue of the $O_{\mathbb{P}}(n^{-1})$ convergence rate of the long-run parameter estimator in Cho et al. (2015, theorem 2). The additional $h^{3/2}$ factor is introduced from the modal bandwidth. In addition, each of $\hat{\alpha}_n$, $\hat{\delta}_n$, $\hat{\gamma}_n$, and $\hat{\phi}_n$ converges at the rate $(nh^3)^{-1/2}$ by Lemma A.5 (iii).
- (ii) Estimating the density function by the smoothing kernel introduces a smoothing bias, which is computed as $b_h := -\mu_2 h^2 f_u^{(3)}(0)/(2f_u^{(2)}(0))$; see the proof of Lemma A.6 (iii). Therefore, $b_h = O(h^2)$, which is also $O((nh^3)^{-1/2})$ if $nh^7 = O(1)$. By this, the smoothing bias can influence the asymptotic distributions of $\hat{\rho}_n$ and $\hat{\alpha}_n$. Lemma A.6 and Theorem 1 given below examine the influence of the

smoothing bias to the asymptotic distributions. $\square$

Let $\mathbf{e}_1$ denote the first standard basis vector, viz., $\mathbf{e}_1 = (1, \mathbf{0}'_{(kq+p) \times 1})'$.

Lemma A.6. *Given Assumption 1,*

(i)

$$\frac{1}{nh^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \mathbf{j}_t \mathbf{j}_t' \xrightarrow{\mathbb{P}} f_u^{(2)}(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}_t'],$$

$$\frac{1}{n^2 h^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \rightsquigarrow f_u^{(2)}(0) \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w', \quad \text{and} \quad \frac{1}{n^{3/2} h^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \mathbf{j}_t \tilde{\mathbf{x}}_t' \xrightarrow{\mathbb{P}} \mathbf{0};$$

$$(ii) \quad n\sqrt{h^3} (\hat{\boldsymbol{\kappa}}_n - \boldsymbol{\beta}_*) \rightsquigarrow \boldsymbol{\ell}_\kappa \sim N\left(\mathbf{0}, \omega_* \left[\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w' \right]^{-1}\right) \text{ conditional on } \tilde{\mathbf{B}}_w(\cdot);$$

$$(iii) \quad \sqrt{nh^3} (\hat{\boldsymbol{\rho}}_n - \boldsymbol{\rho}_\dagger - b_h \mathbf{e}_1) \rightsquigarrow \boldsymbol{\ell}_\rho - \mathbf{e}_1 \int \tilde{\mathbf{B}}_w' \boldsymbol{\ell}_\kappa \text{ such that}$$

$$\left(\frac{1}{nh^3} \sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \mathbf{j}_t' \right)^{-1} \left(\frac{1}{\sqrt{nh}} \sum_{t=1}^n \bar{\phi}_t^{(1)} \mathbf{j}_t \right) \rightsquigarrow \boldsymbol{\ell}_\rho \sim N(\mathbf{0}, \boldsymbol{\Psi}^+);$$

(iv) $\boldsymbol{\ell}_\rho$ and $\boldsymbol{\ell}_\kappa$ are independent;

$$(v) \quad \sqrt{nh^3} ((\hat{\mathbf{v}}_n - \mathbf{v}_\dagger)', (\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_*)')' \overset{A}{\rightsquigarrow} N(\mathbf{0}, \boldsymbol{\Psi}). \quad \square$$

Remark A.4. Lemma A.6 (iv) implies that the weak limit of the slope coefficient given in Lemma A.6 (v) is independent of $\boldsymbol{\ell}_\kappa$. $\square$

Let $\ddot{\mathbf{x}}_t'$ be the t -th row vector of $\ddot{\mathbf{X}} := \mathbf{P}_J \mathbf{X}$, where $\mathbf{P}_J := (\mathbf{I} - \mathbf{J}(\mathbf{J}'\mathbf{J})^{-1}\mathbf{J}')$ such that $\mathbf{J} := [\mathbf{j}_1, \dots, \mathbf{j}_n]'$. We also let $\mathbf{s}_\alpha = (1, \mathbf{0}'_{kq}, -\mu_* \boldsymbol{\iota}_p')'$.

Lemma A.7. *Given Assumption 1,*

$$(i) \quad (i.a) \quad (n^2 h^3)^{-1} \sum_{t=1}^n \phi_t^{(2)} (\ddot{\mathbf{x}}_t \ddot{\mathbf{x}}_t' - \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t') = o_{\mathbb{P}}(1);$$

$$(i.b) \quad (n\sqrt{h})^{-1} \sum_{t=1}^n \phi_t^{(1)} (\ddot{\mathbf{x}}_t - \tilde{\mathbf{x}}_t) = o_{\mathbb{P}}(1);$$

$$(ii) \quad (n^{3/2} h^3)^{-1} \sum_{t=1}^n \phi_t^{(2)} \mathbf{x}_t' \rightsquigarrow f_u^{(2)}(0) \int \bar{\mathbf{B}}_w';$$

$$(iii) \quad \sum_{t=1}^n \phi_t^{(2)} \bar{\mathbf{w}}_t \mathbf{x}_t' = O_{\mathbb{P}}(n\sqrt{h});$$

$$(iv) \quad \sum_{t=1}^n \phi_t^{(2)} \mathbf{k}_t \mathbf{x}_t' = O_{\mathbb{P}}(n^{3/2} h^3);$$

$$(v) \quad \sqrt{nh^3} (\hat{\alpha}_n - \alpha_* - b_h) \rightsquigarrow \mathbf{s}'_\alpha \boldsymbol{\ell}_\rho - \left(\int \bar{\mathbf{B}}_w' \right) \boldsymbol{\ell}_\kappa;$$

$$(vi) \quad (n^{3/2} h^3)^{-1} \sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \tilde{\mathbf{x}}_t' = o_{\mathbb{P}}(1). \quad \square$$

Remark A.5. (i) The smoothing bias b_h is associated with only the intercept estimation when $f_u^{(3)}(0) \neq 0$.

This bias vanishes if the undersmoothing condition holds: $nh^7 \rightarrow 0$. However, the bandwidth for optimizing AMSE does not satisfy the undersmoothing condition as pointed out by Yao and Li (2014).

If $h \propto n^{-1/7}$, then it holds that $nh^5 \rightarrow \infty$ for the consistency of the estimator, but it does not satisfy $nh^7 \rightarrow 0$ for bias elimination.

- (ii) Lemma A.6 (ii) states that the long-run direction is estimated at the rate $n\sqrt{h^3}$ with a mixed normal limit, and both features have a simple source. The rate is faster than the stationary rate $\sqrt{nh^3}$ by $\sqrt{n}$ because $\sum_{t=1}^n \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' = O_{\mathbb{P}}(n^2)$. The limit is mixed normal because that Hessian does not converge to a constant. That is, $n^{-2} \sum_{t=1}^n \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \rightsquigarrow \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w'$. Conditionally on this limit, the score is Gaussian, so that the ratio is asymptotically mixed normal. This is the modal counterpart of the mixed normal limit for the long-run coefficient in Cho et al. (2015) with ω_* replacing $\tau(1 - \tau)/f_\tau^2$. Because the mixing variate cancels in a standardized statistic, the Wald test in Section 2.2 remains asymptotically chi-squared under the null. $\square$

Proof of Lemma A.1: Assumption 1 (vi) yields that $n^{-1/2} \sum_{t=1}^n \bar{\mathbf{w}}_t \rightsquigarrow \mathbf{B}_w(1)$ and $n^{-1/2} \sum_{t=1}^n \bar{\mathbf{k}}_t \rightsquigarrow \mathbf{B}_k(1)$. Additionally, it follows that $n^{-1} \sum_{t=1}^n \mathbf{k}_t \xrightarrow{\mathbb{P}} \mathbb{E}[\mathbf{k}_t]$ by the ergodic theorem. This completes the proof. $\blacksquare$

Proof of Lemma A.2: (i) Utilizing the ergodic theorem given Assumption 1 (iv), Lemma A.2 (i) follows.

(ii) We follow the proof of Proposition 18.1(d) in Hamilton (2020, pp. 562–563). Specifically, $\mathbf{u}_{t-s}$ and ξ_{t-1} in Hamilton (2020) correspond to $\mathbf{w}_{t-i}$ and $\mathbf{x}_t$ of here. Moreover, by Assumptions 1 (iv) and (vi), $\sum_{t=1}^n \mathbf{w}_{t-i} \mathbf{x}_t' = O_{\mathbb{P}}(n)$ and $n^{-1/2} \sum_{t=1}^{\lfloor n(\cdot) \rfloor} \mathbf{w}_t \rightsquigarrow \bar{\mathbf{B}}_w(\cdot)$. By the continuous mapping theorem and Assumptions 1 (i) and (iv), it follows that $\sum_{t=1}^n u_{t-i} \mathbf{x}_t = O_{\mathbb{P}}(n^{3/2})$. Specifically, $\sum_{t=1}^n u_{t-i} \mathbf{x}_t = \mathbb{E}[u_t] \sum_{t=1}^n \mathbf{x}_t + \sum_{t=1}^n (u_{t-i} - \mathbb{E}[u_t]) \mathbf{x}_t = O_{\mathbb{P}}(n^{3/2}) + O_{\mathbb{P}}(n)$ by Lemma A.3 (i).

(iii)–(v) The proofs follow from the stationarity and ergodicity of $\{k_{t,i}\}$, which is established by Assumption 1 (v) and Brockwell and Davis (1991, theorem 4.4.1) combined with Lemma A.2 (ii) and Assumptions 1 (iv) and (vi). The detailed arguments are identical to the proof of Lemma A2 in Cho et al. (2015). $\blacksquare$

Proof of Lemma A.3: (i and ii) By Assumptions 1 (iv) and (vi), $n^{-1/2} \sum_{t=1}^{\lfloor n(\cdot) \rfloor} \mathbf{w}_t \rightsquigarrow \bar{\mathbf{B}}_w(\cdot)$, and this implies that $n^{-1/2} \mathbf{x}_{\lfloor nr \rfloor} = n^{-1/2} \sum_{t=1}^{\lfloor nr \rfloor} \mathbf{w}_t \rightsquigarrow \bar{\mathbf{B}}_w(r)$ with $n^{-1/2} \mathbf{x}_0 = o_{\mathbb{P}}(1)$. We next apply the continuous mapping theorem and Lemma 3.1(e) of Phillips and Durlauf (1986, lemma 3.1.e) to obtain the desired results. $\blacksquare$

Proof of Lemma A.4: (i) By Lemmas A.1 (i), A.2 (i) and (ii), it follows that $\{n^{-1} \sum_{t=1}^n \bar{\mathbf{w}}_t, n^{-1} \sum_{t=1}^n \bar{\mathbf{w}}_t \bar{\mathbf{w}}_t', n^{-3/2} \sum_{t=1}^n \bar{\mathbf{w}}_t \mathbf{x}_t'\} \xrightarrow{\mathbb{P}} \{\mathbf{0}, \mathbb{E}[\bar{\mathbf{w}}_t \bar{\mathbf{w}}_t'], \mathbf{0}\}$. Furthermore, Lemma A.3 provides the limits of $n^{-3/2} \sum_{t=1}^n \mathbf{x}_t$ and $n^{-2} \sum_{t=1}^n \mathbf{x}_t \mathbf{x}_t'$. All the convergences are jointly achieved because each of them is a continuous functional of $\mathbf{B}_n(\cdot)$, whose weak limit is given in Assumption 1 (vi).

(ii) The first two components are equal to the third and fourth components of $B_n(1)$ in Assumption 1 (vi). For the final component, we let $\varepsilon_{nt} := \bar{\phi}^{(1)}(u_{th})/\sqrt{h}$, noting that $\{\varepsilon_{nt}, \mathcal{F}_t\}$ is an MDA with $\sup_{n,t} \mathbb{E}|\varepsilon_{nt}|^{2+\lambda} < \infty$ by Assumption 1 (ii.b) and (iii.a). Furthermore, Assumption 1 (vi) implies that $n^{-1/2}(\mathbf{x}_{[n(\cdot)]}, \sum_{t \leq [n(\cdot)]} \varepsilon_{nt}) \rightsquigarrow (\bar{\mathcal{B}}_w(\cdot), \mathcal{B}_\phi(\cdot))$, so that $n^{-1} \sum_{t=1}^n (n^{-1/2} \mathbf{x}_t) \varepsilon_{nt} \rightsquigarrow \int \bar{\mathcal{B}}_w d\mathcal{B}_\phi$ by Hansen (1992, theorem 3.1). Here, the contribution of $\mathbf{x}_0$ is asymptotically negligible, because $n^{-1} h^{-1/2} \mathbf{x}_0 \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) = \mathbf{x}_0 O_{\mathbb{P}}(n^{-1/2}) = o_{\mathbb{P}}(1)$.

(iii) We first note that $D_H^{-1} G' \mathbf{k} = [n^{-1} \sum_{t=1}^n \mathbf{k}_t, n^{-1} \sum_{t=1}^n (\bar{\mathbf{w}}_t \mathbf{k}_t')', n^{-3/2} \sum_{t=1}^n (\mathbf{x}_t \mathbf{k}_t')']'$, and each element is $O_{\mathbb{P}}(1)$ by Lemmas A.1 (ii), A.2 (iv), and A.2 (v), respectively. Therefore, $D_H^{-1} G' \mathbf{k} = O_{\mathbb{P}}(1)$.

(iv) We first note that

$$n^{-2} \mathbf{X}' P_{\widehat{\mathbf{W}}} \mathbf{X} = n^{-2} \mathbf{X}' \mathbf{X} - (n^{-3/2} \mathbf{X}' \widehat{\mathbf{W}}) (n^{-1} \widehat{\mathbf{W}}' \widehat{\mathbf{W}})^{-1} (n^{-3/2} \widehat{\mathbf{W}}' \mathbf{X}).$$

Given this, Lemma A.3 (ii) implies that $n^{-2} \mathbf{X}' \mathbf{X} \rightsquigarrow \int \bar{\mathcal{B}}_w \bar{\mathcal{B}}_w'$. Furthermore, we note that $n^{-1} \widehat{\mathbf{W}}' \widehat{\mathbf{W}} \xrightarrow{\mathbb{P}} \text{diag}(1, \mathbb{E}[\bar{\mathbf{w}}_t \bar{\mathbf{w}}_t'])$ and $n^{-3/2} \widehat{\mathbf{W}}' \mathbf{X} \rightsquigarrow [\int \bar{\mathcal{B}}_w, \mathbf{0}]'$ by (i), so that

$$n^{-2} \mathbf{X}' P_{\widehat{\mathbf{W}}} \mathbf{X} \rightsquigarrow \int \bar{\mathcal{B}}_w \bar{\mathcal{B}}_w' - \int \bar{\mathcal{B}}_w \int \bar{\mathcal{B}}_w' = \int \tilde{\mathcal{B}}_w \tilde{\mathcal{B}}_w'.$$

(v) We note that $\sum_{t=1}^n \phi^{(1)}(u_{th}) \tilde{\mathbf{x}}_t = \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \tilde{\mathbf{x}}_t$, because the rows of $P_{\widehat{\mathbf{W}}} \mathbf{X}$ sum to zero. Therefore,

$$\begin{aligned} \frac{1}{n\sqrt{h}} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \tilde{\mathbf{x}}_t &= \frac{1}{n\sqrt{h}} \mathbf{X}' \bar{\Phi}^{(1)}(\mathbf{u}_h) - (n^{-3/2} \mathbf{X}' \widehat{\mathbf{W}}) (n^{-1} \widehat{\mathbf{W}}' \widehat{\mathbf{W}})^{-1} \left(\frac{1}{\sqrt{nh}} \widehat{\mathbf{W}}' \bar{\Phi}^{(1)}(\mathbf{u}_h) \right) \\ &\rightsquigarrow \int \bar{\mathcal{B}}_w d\mathcal{B}_\phi - \left[\int \bar{\mathcal{B}}_w, \mathbf{0} \right] \text{diag}[1, \mathbb{E}[\bar{\mathbf{w}}_t \bar{\mathbf{w}}_t']] \begin{bmatrix} \mathcal{B}_\phi(1) \\ \mathcal{B}_{\phi \cdot \mathbf{w}}(1) \end{bmatrix} \\ &= \int \bar{\mathcal{B}}_w d\mathcal{B}_\phi - \int \bar{\mathcal{B}}_w \mathcal{B}_\phi(1) \end{aligned}$$

by (i) and (ii). We further note that $\int \bar{\mathcal{B}}_w d\mathcal{B}_\phi - \int \bar{\mathcal{B}}_w \cdot \mathcal{B}_\phi(1) = \int \tilde{\mathcal{B}}_w d\mathcal{B}_\phi$ using (10).

(vi) Given that $\mathbf{j}_t := (1, \bar{\mathbf{w}}_t', \mathbf{k}_t')'$, $(nh)^{-1/2} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \mathbf{j}_t$ is a linear map of the $\mathcal{B}_\phi$ -, $\mathcal{B}_{\phi \cdot \mathbf{w}}$ -, and $\mathcal{B}_{\phi \cdot \mathbf{k}}$ -components of $B_n(1)$, so that it is asymptotically normal by Assumption 1 (vi). Furthermore, its covariance matrix converges to $v^2 f_u(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}_t']$ because $\mathbb{E}[\{\bar{\phi}^{(1)}(u_{th})\}^2 | \mathcal{F}_{t-1}]$ is constant under Assumptions (i.b) and (iv.b), so that $h^{-1} \mathbb{E}[\{\bar{\phi}^{(1)}(u_{th})\}^2 \mathbf{j}_t \mathbf{j}_t'] = h^{-1} \mathbb{E}[\{\bar{\phi}^{(1)}(u_{th})\}^2] \mathbb{E}[\mathbf{j}_t \mathbf{j}_t'] \rightarrow v^2 f_u(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}_t']$ by (A.3). This completes the proof. $\blacksquare$

Proof of Lemma A.5: If we let $\boldsymbol{\vartheta} := (\boldsymbol{\rho}', \boldsymbol{\kappa}')'$ and $\tilde{\mathbf{z}}_t := (\mathbf{j}_t', \mathbf{x}_t')'$, then $\mathbf{z}_t' \boldsymbol{\alpha}_* = \tilde{\mathbf{z}}_t' \boldsymbol{\vartheta}_\dagger$ by (12). Furthermore,

from the definitions of $\varsigma := \boldsymbol{\lambda} + \boldsymbol{\Gamma}_* \boldsymbol{\phi}$ and $\boldsymbol{\rho} := (\nu, \boldsymbol{v}', \boldsymbol{\phi}')'$, there exists a linear bijection relationship between $\boldsymbol{\alpha}$ and $\boldsymbol{\vartheta}$. Using this fact, we let $\tilde{Q}_{nh}(\cdot)$ be a function defined on the space of $\boldsymbol{\vartheta}$ such that for each $\boldsymbol{\alpha}$, $Q_{nh}(\boldsymbol{\alpha}) = \tilde{Q}_{nh}(\boldsymbol{\vartheta})$.

To show the given claims, we let $\tilde{\boldsymbol{D}}_n := \text{diag}(\sqrt{nh^3} \boldsymbol{I}_{1+kq+p}, n\sqrt{h^3} \boldsymbol{I}_k)$ and show that for any $\eta > 0$ there exists $\theta < \infty$ such that

$$\mathbb{P} \left\{ \sup_{\|\boldsymbol{c}\|=\theta} \tilde{Q}_{nh}(\boldsymbol{\vartheta}_{\dagger} + \tilde{\boldsymbol{D}}_n^{-1} \boldsymbol{c}) < \tilde{Q}_{nh}(\boldsymbol{\vartheta}_{\dagger}) \right\} \geq 1 - \eta, \quad (\text{A.7})$$

following [Fan and Li \(2001\)](#). For this we approximate $\tilde{Q}_{nh}(\boldsymbol{\vartheta}_{\dagger} + \tilde{\boldsymbol{D}}_n^{-1} \boldsymbol{c})$ by a third-order expansion as follows:

$$\tilde{Q}_{nh}(\boldsymbol{\vartheta}_{\dagger} + \tilde{\boldsymbol{D}}_n^{-1} \boldsymbol{c}) - \tilde{Q}_{nh}(\boldsymbol{\vartheta}_{\dagger}) = I_1 + I_2 + I_3,$$

where

$$\begin{aligned} I_1 &:= -\frac{1}{nh^2} \sum_{t=1}^n \phi^{(1)}(u_{th}) d_{nt}(\boldsymbol{c}), \quad I_2 := \frac{1}{2nh^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) (d_{nt}(\boldsymbol{c}))^2, \quad \text{and} \\ I_3 &:= -\frac{1}{2nh^4} \sum_{t=1}^n d_{nt}(\boldsymbol{c})^3 \int_0^1 (1-s)^2 \phi^{(3)}\left(u_{th} - \frac{s}{h} d_{nt}(\boldsymbol{c})\right) ds \quad \text{such that} \quad d_{nt}(\boldsymbol{c}) := \boldsymbol{c}' \tilde{\boldsymbol{D}}_n^{-1} \tilde{\boldsymbol{z}}_t. \end{aligned}$$

We examine each term on the right side. First, we examine I_1 . For this we decompose $\phi^{(1)}(u_{th}) = \bar{\phi}^{(1)}(u_{th}) + m_h$ with $m_h = O(h^4)$. Therefore,

$$I_1 = -\frac{1}{nh^2} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) d_{nt}(\boldsymbol{c}) + \frac{1}{nh^2} \sum_{t=1}^n m_h d_{nt}(\boldsymbol{c}).$$

Here, we note that $\tilde{\boldsymbol{z}}_t = (\boldsymbol{j}'_t, \boldsymbol{x}'_t)'$ and that for given $\boldsymbol{c}$,

$$\frac{1}{nh^2} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) d_{nt}(\boldsymbol{c}) = \frac{1}{nh^2} \boldsymbol{c}' \tilde{\boldsymbol{D}}_n^{-1} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) (\boldsymbol{j}'_t, \boldsymbol{x}'_t)' = O_{\mathbb{P}}((nh^3)^{-1}),$$

because $\sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \boldsymbol{j}_t = O_{\mathbb{P}}(\sqrt{nh})$ and $\sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \boldsymbol{x}_t = O_{\mathbb{P}}(n\sqrt{h})$ by [Lemma A.4 \(ii\)](#). In addition, we note that $\tilde{\boldsymbol{z}}_t = (\boldsymbol{j}'_t, \boldsymbol{x}'_t)' = (1, \bar{\boldsymbol{w}}'_t, \boldsymbol{k}'_t, \boldsymbol{x}'_t)'$, so that

$$\frac{1}{nh^2} \sum_{t=1}^n m_h d_{nt}(\boldsymbol{c}) = \frac{m_h}{nh^2} \boldsymbol{c}' \tilde{\boldsymbol{D}}_n^{-1} \sum_{t=1}^n (1, \bar{\boldsymbol{w}}'_t, \boldsymbol{k}'_t, \boldsymbol{x}'_t)' = O_{\mathbb{P}}((n^{-1}h)^{1/2}),$$

because $m_h = O(h^4)$, $\sum_{t=1}^n 1 = O(n)$, $\sum_{t=1}^n \bar{\boldsymbol{w}}_t = O_{\mathbb{P}}(\sqrt{n})$, $\sum_{t=1}^n \boldsymbol{k}_t = O_{\mathbb{P}}(n)$, and $\sum_{t=1}^n \boldsymbol{x}_t = O_{\mathbb{P}}(n^{3/2})$ by [Lemma A.1](#). We further note that $(h/n)^{1/2} = O((nh^3)^{-1}) + o_{\mathbb{P}}(1)$, given that $nh^7 = O(1)$.

Therefore, $I_1 = O_{\mathbb{P}}((nh^3)^{-1})$.

Second, we examine I_2 . We first note that

$$I_2 = \frac{1}{2nh^3} \sum_{t=1}^n \left\{ \left[\phi^{(2)}(u_{th}) - \mathbb{E}[\phi^{(2)}(u_{th})|\mathcal{F}_{t-1}] \right] + \mathbb{E}[\phi^{(2)}(u_{th})|\mathcal{F}_{t-1}] \right\} (d_{nt}(\mathbf{c}))^2.$$

Because u_t is independent of $\mathcal{F}_{t-1}$, it follows that $\mathbb{E}[\phi^{(2)}(u_{th})|\mathcal{F}_{t-1}] = h^3 f_u^{(2)}(0)\{1 + o(h)\}$ uniformly by (A.5). Therefore,

$$\begin{aligned} \frac{1}{2nh^3} \sum_{t=1}^n \mathbb{E}[\phi^{(2)}(u_{th})|\mathcal{F}_{t-1}](d_{nt}(\mathbf{c}))^2 &= \frac{f_u^{(2)}(0)}{2} \frac{1}{n} \sum_{t=1}^n (d_{nt}(\mathbf{c}))^2 \{1 + o_{\mathbb{P}}(1)\} \\ &= \frac{f_u^{(2)}(0)}{2nh^3} \mathbf{c}' \ddot{D}_n^{-1} \left(\sum_{t=1}^n \tilde{z}_t \tilde{z}_t' \right) \ddot{D}_n^{-1} \mathbf{c} \{1 + o_{\mathbb{P}}(1)\}, \end{aligned}$$

where $\ddot{D}_n := h^{-3/2} \tilde{D}_n$. We further note that

$$M_n := \ddot{D}_n^{-1} \left(\sum_{t=1}^n \tilde{z}_t \tilde{z}_t' \right) \ddot{D}_n^{-1} \rightsquigarrow \mathcal{M} := \begin{bmatrix} \mathbb{E}[\mathbf{j}_t \mathbf{j}_t'] & \mathbb{E}[\mathbf{j}_t] \int \bar{\mathbf{B}}_w' \\ \int \bar{\mathbf{B}}_w \mathbb{E}[\mathbf{j}_t]' & \int \bar{\mathbf{B}}_w \bar{\mathbf{B}}_w' \end{bmatrix}$$

by Lemmas A.1, A.2 (i, iii, iv), A.3, and A.4 (i). Because $\mathbb{E}[\mathbf{j}_t]$ is the first column of $\mathbb{E}[\mathbf{j}_t \mathbf{j}_t']$, the Schur complement of $\mathcal{M}$ is $\int \bar{\mathbf{B}}_w \bar{\mathbf{B}}_w' - \int \bar{\mathbf{B}}_w \mathbb{E}[\mathbf{j}_t]' = \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w'$, which is positive definite a.s. by Assumption 1 (iv.c) and (vi), so that $\mathcal{M}$ is positive definite with probability 1.

Next, we examine the MDA part of I_2 . If we write $\bar{\phi}_t^{(2)} := \phi^{(2)}(u_{th}) - \mathbb{E}[\phi^{(2)}(u_{th})|\mathcal{F}_{t-1}]$, then $\{\bar{\phi}_t^{(2)}(d_{nt}(\mathbf{c}))^2, \mathcal{F}_t\}$ is an MDA, because $(d_{nt}(\mathbf{c}))^2$ is $\mathcal{F}_{t-1}$ -measurable. By (A.6), $\mathbb{E}[(\bar{\phi}_t^{(2)})^2] = O(h)$, so that

$$\text{var} \left(\frac{1}{2nh^3} \sum_{t=1}^n \bar{\phi}_t^{(2)}(d_{nt}(\mathbf{c}))^2 \right) = \frac{1}{4n^2 h^6} \sum_{t=1}^n \mathbb{E}[(\bar{\phi}_t^{(2)})^2 (d_{nt}(\mathbf{c}))^4] = \frac{O(h)}{4n^2 h^6} \sum_{t=1}^n \mathbb{E}[(d_{nt}(\mathbf{c}))^4].$$

For this calculation, if we first let $\xi_n := (nh^3)^{-1/2}$ and $\mathbf{r}_{nt} := (\mathbf{j}_t', n^{-1/2} \mathbf{x}_t')'$, then $\tilde{D}_n^{-1} \tilde{z}_t = \xi_n \mathbf{r}_{nt}$, and for $\ell = 3, 4, 6$,

$$\frac{1}{n} \sum_{t=1}^n \mathbb{E}|\mathbf{c}' \mathbf{r}_{nt}|^\ell = O(\theta^\ell) \tag{A.8}$$

uniformly over $\mathbf{c}$ such that $\|\mathbf{c}\| = \theta$ by Assumption 1 (iv.d) and $\mathbb{E}\|\mathbf{x}_t\|^6 = O(t^3)$. Therefore,

$$\sum_{t=1}^n \mathbb{E}[(d_{nt}(\mathbf{c}))^4] = \xi_n^4 \sum_{t=1}^n \mathbb{E}[(\mathbf{c}' \mathbf{r}_{nt})^4] = O\left(\frac{\theta^4}{nh^6}\right)$$

by noting $d_{nt}(\mathbf{c}) = \xi_n \mathbf{c}' \mathbf{r}_{nt}$. We now substitute this into the preceding variance, letting the variance be of order $O(\theta^4/(n^3 h^{11}))$. From this, the order of the standard deviation is of $O(\theta^2/(n^{3/2} h^{11/2}))$. Since the systematic quadratic term is of order $O_{\mathbb{P}}(\theta^2/(n h^3))$, their ratio is $(n h^5)^{-1/2} = o(1)$. Thus the centered part of I_2 is $o_{\mathbb{P}}(\theta^2/(n h^3))$.

We now combine these two results. If we recall that $f_u^{(2)}(0) < 0$ from Assumption 1 (iii.c), it follows that

$$I_2 = \frac{f_u^{(2)}(0)}{2n h^3} \mathbf{c}' \mathbf{M}_n \mathbf{c} \{1 + o_{\mathbb{P}}(1)\} \leq -\frac{\kappa_n \theta^2}{(n h^3)},$$

on a set whose probability tends to one, where $\kappa_n := \frac{1}{2} |f_u^{(2)}(0)| \lambda_{\min}(\mathbf{M}_n) \{1 + o_{\mathbb{P}}(1)\}$, $\lambda_{\min}(\cdot)$ denotes the smallest eigenvalue, and $\|\mathbf{c}\| = \theta$. As $\mathbf{M}_n \rightsquigarrow \mathbf{M}$ with $\mathbf{M}$ being positive definite with probability one, $\lambda_{\min}(\mathbf{M}_n)$ is bounded away from zero in probability, and so is κ_n .

Third, we examine I_3 . For fixed $s \in [0, 1]$, $d_{nt}(\mathbf{c})$ is $\mathcal{F}_{t-1}$ -measurable. Given this, we first apply integration by parts three times consecutively to obtain

$$\begin{aligned} \mathbb{E} \left[\phi^{(3)} \left(u_{th} - \frac{s}{h} d_{nt}(\mathbf{c}) \right) \middle| \mathcal{F}_{t-1} \right] &= h \int \phi^{(3)}(v) f_u \{h v + s d_{nt}(\mathbf{c})\} dv \\ &= -h^4 \int \phi(v) f_u^{(3)} \{h v + s d_{nt}(\mathbf{c})\} dv = O(h^4) \end{aligned}$$

uniformly in s, t , and $\mathbf{c}$ such that $\|\mathbf{c}\| = \theta$ by Assumptions 1 (ii.a) and (iii.a). Meanwhile, the corresponding conditional second moment is bounded by

$$h \|f_u\|_{\infty} \int [\phi^{(3)}(v)]^2 dv = O(h). \quad (\text{A.9})$$

We now decompose the integrand in I_3 into its conditional mean and its MDA. By (A.14), the order of the conditional-mean part is of

$$O_{\mathbb{P}} \left(\frac{h^4}{n h^4} \sum_{t=1}^n |d_{nt}(\mathbf{c})|^3 \right) = O_{\mathbb{P}}(\theta^3 \xi_n^3) = o_{\mathbb{P}} \left(\frac{\theta^2}{n h^3} \right),$$

because $\xi_n \rightarrow 0$. Given this, the order of standard deviation of the martingale part is of

$$O_{\mathbb{P}} \left(\frac{1}{n h^4} \left\{ h \sum_{t=1}^n |d_{nt}(\mathbf{c})|^6 \right\}^{1/2} \right) = O_{\mathbb{P}} \left(\frac{\theta^3}{n^2 h^8} \right) = o_{\mathbb{P}} \left(\frac{\theta^2}{n h^3} \right)$$

by the conditional orthogonality, (A.9), and (A.8), where the last equality follows by $n h^5 \rightarrow \infty$. Since

the parameter space is finite dimensional, boundedness of $\phi^{(4)}(\cdot)$, and the sixth finite moment bound yield stochastic equicontinuity over the compact sphere by the usual finite-covering argument. Consequently,

$$I_3 = o_{\mathbb{P}}\left(\frac{\theta^2}{nh^3}\right)$$

uniformly over c such that $\|c\| = \theta$.

Finally, if we combine the three terms, (A.7) holds for sufficiently large θ , which delivers a local maximizer with $\|\tilde{D}_n(\hat{\boldsymbol{\vartheta}}_n - \boldsymbol{\vartheta}_*)\| = O_{\mathbb{P}}(1)$, which proves (i) and (ii). Part (iii) follows from $\hat{\boldsymbol{\lambda}}_n = \hat{\boldsymbol{\varsigma}}_n - \boldsymbol{\Gamma}_* \hat{\boldsymbol{\phi}}_n$. In particular, $\hat{\boldsymbol{\gamma}}_n - \boldsymbol{\gamma}_* = (\hat{\boldsymbol{\kappa}}_n - \boldsymbol{\beta}_*) - \boldsymbol{\beta}_* \boldsymbol{\nu}'_p(\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_*) = O_{\mathbb{P}}((nh^3)^{-1/2})$. This completes the proof. $\blacksquare$

Proof of Lemma A.6: Throughout this proof, we write $\phi_t^{(j)} := \phi^{(j)}(u_{th})$ for notational simplicity and work with the reparameterized model in (12) so that the objective function $Q_{nh}(\cdot)$ is translated into $\tilde{Q}_{nh}(\cdot)$ such that the parameters and regressors are given as $\boldsymbol{\vartheta} = (\boldsymbol{\rho}', \boldsymbol{\kappa}')'$ and $\tilde{\mathbf{z}}_t = (\mathbf{j}'_t, \mathbf{x}'_t)'$ with $\mathbf{j}_t = (1, \bar{\mathbf{w}}'_t, \mathbf{k}'_t)'$, respectively. By (13), $\boldsymbol{\kappa}_{\dagger} = \boldsymbol{\beta}_*$, so that $\hat{\boldsymbol{\kappa}}_n$ estimates $\boldsymbol{\beta}_*$. Given this, the first-order condition for $\hat{\boldsymbol{\vartheta}}_n$ is given as $\sum_{t=1}^n \phi_t^{(1)}((y_t - \tilde{\mathbf{z}}'_t \hat{\boldsymbol{\vartheta}}_n)/h) \tilde{\mathbf{z}}_t \equiv \mathbf{0}$, and we expand it around $\boldsymbol{\vartheta}_{\dagger}$, obtaining the following block system:

$$\mathbf{s}_j = h^{-1}[\mathbf{T}_{jj} \mathbf{a}_n + \mathbf{T}_{jx} \mathbf{b}_n] + o_{\mathbb{P}}(\sqrt{nh}) \quad \text{and} \quad \mathbf{s}_x = h^{-1}[\mathbf{T}_{xj} \mathbf{a}_n + \mathbf{T}_{xx} \mathbf{b}_n] + o_{\mathbb{P}}(n\sqrt{h}), \quad (\text{A.10})$$

where

$$\begin{aligned} \mathbf{a}_n &:= \hat{\boldsymbol{\rho}}_n - \boldsymbol{\rho}_{\dagger}, & \mathbf{b}_n &:= \hat{\boldsymbol{\kappa}}_n - \boldsymbol{\kappa}_{\dagger}, & \mathbf{s}_j &:= \sum_{t=1}^n \phi_t^{(1)} \mathbf{j}_t, & \mathbf{s}_x &:= \sum_{t=1}^n \phi_t^{(1)} \mathbf{x}_t, \\ \mathbf{T}_{jj} &:= \sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \mathbf{j}'_t, & \mathbf{T}_{jx} &:= \sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \mathbf{x}'_t = \mathbf{T}'_{xj}, & \text{and} & & \mathbf{T}_{xx} &:= \sum_{t=1}^n \phi_t^{(2)} \mathbf{x}_t \mathbf{x}'_t. \end{aligned}$$

(i) We first examine the asymptotic behavior of $\mathbf{T}_{jj}$. For this we first note that $\mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}] = h^3 f_u^{(2)}(0) \{1 + o(h)\}$ by (A.5). Therefore,

$$\frac{1}{nh^3} \mathbf{T}_{jj} = \frac{1}{nh^3} \sum_{t=1}^n (\phi_t^{(2)} - \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}]) \mathbf{j}_t \mathbf{j}'_t + \frac{1}{nh^3} \sum_{t=1}^n \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}] \mathbf{j}_t \mathbf{j}'_t.$$

Here, the second term converges to $f_u^{(2)}(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}'_t]$ by the ergodic theorem, and the first term is $o_{\mathbb{P}}(1)$. Specifically, if we let $\bar{\phi}_t^{(2)} := \phi_t^{(2)} - \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}]$, then the following holds. First, $\{\bar{\phi}_t^{(2)}, \mathcal{F}_t\}$ is an MDA by construction, and $\mathbb{E}[(\bar{\phi}_t^{(2)})^2] = O(h)$ by (A.6). Second, $\mathbf{j}_t$ is $\mathcal{F}_{t-1}$ -measurable, so that $\{\bar{\phi}_t^{(2)} \mathbf{j}_t \mathbf{j}'_t, \mathcal{F}_t\}$ is also

an MDA. Third,

$$\text{var}\left(\sum_{t=1}^n \bar{\phi}_t^{(2)} \mathbf{j}_t \mathbf{j}_t'\right) = \sum_{t=1}^n \mathbb{E}[(\bar{\phi}_t^{(2)})^2 \mathbf{j}_t \mathbf{j}_t' \otimes \mathbf{j}_t \mathbf{j}_t'] = O(h) n \mathbb{E}[\|\mathbf{j}_t\|^4] = O(nh),$$

where $\mathbb{E}\|\mathbf{j}_t\|^4 < \infty$ by Assumption 1 (iv.d). Fourth, $\sum_{t=1}^n \bar{\phi}_t^{(2)} \mathbf{j}_t \mathbf{j}_t' = O_{\mathbb{P}}(\sqrt{nh})$ by applying Chebyshev's inequality, so that if we divide it by nh^3 , then the first term is $O_{\mathbb{P}}(\sqrt{nh}/(nh^3)) = O_{\mathbb{P}}((nh^5)^{-1/2}) = o_{\mathbb{P}}(1)$ as $nh^5 \rightarrow \infty$.

We next examine the asymptotic behavior of $\sum_{t=1}^n \phi_t^{(2)} \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t'$. In parallel to $\mathbf{T}_{jj}$,

$$\begin{aligned} \frac{1}{n^2 h^3} \sum_{t=1}^n \phi_t^{(2)} \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' &= \frac{1}{n^2 h^3} \sum_{t=1}^n (\phi_t^{(2)} - \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}]) \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' + \frac{1}{n^2 h^3} \sum_{t=1}^n \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}] \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \\ &= \frac{1}{n^2 h^3} \sum_{t=1}^n (\phi_t^{(2)} - \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}]) \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' + \frac{f_u^{(2)}(0)}{n^2} \sum_{t=1}^n \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \{1 + o_{\mathbb{P}}(1)\}. \end{aligned}$$

Here, the second term on the right side weakly converges to $f_u^{(2)}(0) \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w'$ by Lemma A.4 (iv). For the first term, $\{\bar{\phi}_t^{(2)} \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t', \mathcal{F}_t\}$ is an MDA because $\tilde{\mathbf{x}}_t$ is $\mathcal{F}_{t-1}$ -measurable, and $\mathbb{E}\|\tilde{\mathbf{x}}_t\|^4 = O(t^2)$ since $\tilde{\mathbf{x}}_t$ is a linear combination of $I(1)$ variables. Hence its variance is $O(h) \sum_{t=1}^n \mathbb{E}\|\tilde{\mathbf{x}}_t\|^4 = O(hn^3)$, so that the sum is $O_{\mathbb{P}}(n^{3/2} \sqrt{h})$ and, after dividing it by $n^2 h^3$, the ratio is $O_{\mathbb{P}}(n^{-1/2} h^{-5/2}) = O_{\mathbb{P}}((nh^5)^{-1/2}) = o_{\mathbb{P}}(1)$.

Finally, we examine the asymptotic behavior of $\sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \tilde{\mathbf{x}}_t'$. As before, we first obtain that

$$\begin{aligned} \frac{1}{n^{3/2} h^3} \sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \tilde{\mathbf{x}}_t' &= \frac{1}{n^{3/2} h^3} \sum_{t=1}^n (\phi_t^{(2)} - \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}]) \mathbf{j}_t \tilde{\mathbf{x}}_t' + \frac{1}{n^{3/2} h^3} \sum_{t=1}^n \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}] \mathbf{j}_t \tilde{\mathbf{x}}_t' \\ &= \frac{1}{n^{3/2} h^3} \sum_{t=1}^n (\phi_t^{(2)} - \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}]) \mathbf{j}_t \tilde{\mathbf{x}}_t' + \frac{f_u^{(2)}(0)}{n^{3/2}} \sum_{t=1}^n \mathbf{j}_t \tilde{\mathbf{x}}_t' \{1 + o_{\mathbb{P}}(1)\}. \end{aligned}$$

Here, for the second term, $\tilde{\mathbf{x}}_t$ is orthogonal to $\hat{\mathbf{w}}_t := (1, \bar{\mathbf{w}}_t')'$ by construction, so that it follows that

$$\frac{1}{n^{3/2}} \sum_{t=1}^n \mathbf{j}_t \tilde{\mathbf{x}}_t' = \frac{1}{n^{3/2}} \sum_{t=1}^n \begin{bmatrix} \hat{\mathbf{w}}_t \tilde{\mathbf{x}}_t' \\ \mathbf{k}_t \tilde{\mathbf{x}}_t' \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ n^{-3/2} \sum_{t=1}^n \mathbf{k}_t \tilde{\mathbf{x}}_t' \end{bmatrix} \rightsquigarrow \begin{bmatrix} \mathbf{0} \\ \mathbb{E}[\mathbf{k}_t] \int \tilde{\mathbf{B}}_w' \end{bmatrix}$$

by applying the proof of Lemma A.2 (v). Furthermore, we note that $\int \tilde{\mathbf{B}}_w \equiv \mathbf{0}$ by construction, implying that the second term on the right converges to zero in probability. For the first term, the same argument applies with $\mathbf{j}_t \tilde{\mathbf{x}}_t'$ in place of $\tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t'$. That is, its variance is $O(h) \sum_{t=1}^n \mathbb{E}[\|\mathbf{j}_t\|^2 \|\tilde{\mathbf{x}}_t\|^2] = O(hn^2)$ by the Cauchy–Schwarz inequality together with Assumption 1 (iv.d), so that the sum is $O_{\mathbb{P}}(n\sqrt{h})$ and, after dividing it by $n^{3/2} h^3$, the ratio is $O_{\mathbb{P}}((nh^5)^{-1/2}) = o_{\mathbb{P}}(1)$.

(ii) If we let $\ddot{\mathbf{X}} = \mathbf{X} - \mathbf{J}\mathbf{C}_n$ such that $\mathbf{C}_n := (\mathbf{J}'\mathbf{J})^{-1}\mathbf{J}'\mathbf{X}$, then it follows from (A.10) that

$$\begin{aligned} \mathbf{s}_x - \mathbf{C}_n' \mathbf{x}_j &= \left(\sum_{t=1}^n \phi_t^{(1)} \ddot{\mathbf{x}}_t \right) + o_{\mathbb{P}}(n\sqrt{h}) \\ &= h^{-1} \left(\sum_{t=1}^n \phi_t^{(2)} \ddot{\mathbf{x}}_t \ddot{\mathbf{x}}_t' \right) \mathbf{b}_n + h^{-1} \left(\sum_{t=1}^n \phi_t^{(2)} \ddot{\mathbf{x}}_t \mathbf{j}_t' \right) (\mathbf{a}_n + \mathbf{C}_n \mathbf{b}_n) + o_{\mathbb{P}}(n\sqrt{h}), \end{aligned} \quad (\text{A.11})$$

Here, $\|\mathbf{C}_n\| = O_{\mathbb{P}}(\sqrt{n})$, so that $\|\mathbf{a}_n + \mathbf{C}_n \mathbf{b}_n\| = O_{\mathbb{P}}((nh^3)^{-1/2})$ by Lemma A.5. Furthermore, Lemma A.7

(vi) and the proof of Lemma A.7 (i) imply $\sum_t \phi_t^{(2)} \ddot{\mathbf{x}}_t \mathbf{j}_t' = o_{\mathbb{P}}(n^{3/2}h^3)$, so that

$$h^{-1} \left(\sum_{t=1}^n \phi_t^{(2)} \ddot{\mathbf{x}}_t \mathbf{j}_t' \right) (\mathbf{a}_n + \mathbf{C}_n \mathbf{b}_n) = o_{\mathbb{P}}(n^{3/2}h^3) h^{-1} O_{\mathbb{P}}((nh^3)^{-1/2}) = o_{\mathbb{P}}(n\sqrt{h}).$$

Thus, we obtain from (A.11) that

$$\mathbf{b}_n = h \left(\sum_{t=1}^n \phi_t^{(2)} \ddot{\mathbf{x}}_t \ddot{\mathbf{x}}_t' \right)^{-1} \sum_{t=1}^n \phi_t^{(1)} \ddot{\mathbf{x}}_t + o_{\mathbb{P}}((n\sqrt{h^3})^{-1}).$$

We now apply Lemmas A.7 (i.a and i.b) to replace $\ddot{\mathbf{x}}_t$ with $\tilde{\mathbf{x}}_t$ in the normalized Hessian and score, respectively, leading to

$$n\sqrt{h^3} \mathbf{b}_n = \left(\frac{1}{n^2 h^3} \sum_{t=1}^n \phi_t^{(2)} \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \right)^{-1} \left(\frac{1}{n\sqrt{h}} \sum_{t=1}^n \phi_t^{(1)} \tilde{\mathbf{x}}_t \right) + o_{\mathbb{P}}(1).$$

We further note that $(n^2 h^3)^{-1} \sum_{t=1}^n \phi_t^{(2)} \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \rightsquigarrow f_u^{(2)}(0) \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w'$ by (i), and $(n^2 h)^{-1/2} \sum_{t=1}^n \phi_t^{(1)} \tilde{\mathbf{x}}_t \rightsquigarrow \int \tilde{\mathbf{B}}_w d\mathcal{B}_\phi$ by Lemma A.4 (v) such that $\int \tilde{\mathbf{B}}_w d\mathcal{B}_\phi \sim N(\mathbf{0}, v^2 f_u(0) \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w')$ conditional on $\tilde{\mathbf{B}}_w(\cdot)$. Therefore,

$$n\sqrt{h^3} \mathbf{b}_n \rightsquigarrow \boldsymbol{\ell}_\kappa := \frac{1}{f_u^{(2)}(0)} \left[\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w' \right]^{-1} \int \tilde{\mathbf{B}}_w d\mathcal{B}_\phi \sim N\left(\mathbf{0}, \omega_* \left[\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w' \right]^{-1}\right)$$

conditional on $\tilde{\mathbf{B}}_w(\cdot)$, given that $\omega_* := v^2 f_u(0) / (f_u^{(2)}(0))^2$.

(iii) If we solve for $\mathbf{a}_n$ from (A.10), it follows that

$$\mathbf{a}_n = -\mathbf{T}_{jj}^{-1} \mathbf{T}_{jx} \mathbf{b}_n + h \mathbf{T}_{jj}^{-1} \mathbf{s}_j + o_{\mathbb{P}}((nh^3)^{-1/2}).$$

We examine each term on the right side. First, we note that

$$\mathbf{T}_{j\mathbf{x}} = \begin{bmatrix} \sum_{t=1}^n \phi_t^{(2)} \mathbf{x}'_t \\ \sum_{t=1}^n \phi_t^{(2)} \bar{\mathbf{w}}_t \mathbf{x}'_t \\ \sum_{t=1}^n \phi_t^{(2)} \mathbf{k}_t \mathbf{x}'_t \end{bmatrix} = \begin{bmatrix} O_{\mathbb{P}}(n^{3/2}h^3) \\ O_{\mathbb{P}}(n\sqrt{h}) \\ O_{\mathbb{P}}(n^{3/2}h^3) \end{bmatrix}$$

by Lemmas A.7 (ii, iii, and iv), so that

$$\frac{1}{\sqrt{n}} \mathbf{T}_{jj}^{-1} \mathbf{T}_{j\mathbf{x}} \rightsquigarrow \mathbb{E}[\mathbf{j}_t \mathbf{j}'_t]^{-1} \mathbb{E}[\mathbf{j}_t] \int \bar{\mathbf{B}}'_w = \mathbf{e}_1 \left(\int \bar{\mathbf{B}}'_w \right)$$

as $nh^5 \rightarrow \infty$ and $\mathbb{E}[\mathbf{j}_t \mathbf{j}'_t] \mathbf{e}_1 = \mathbb{E}[\mathbf{j}_t]$ by noting that the first-row element of $\mathbf{j}_1$ is unity. Furthermore, (ii) provides the asymptotic behavior of $\mathbf{b}_n$, so that

$$\sqrt{nh^3} \mathbf{T}_{jj}^{-1} \mathbf{T}_{j\mathbf{x}} \mathbf{b}_n \rightsquigarrow \mathbf{e}_1 \left(\int \bar{\mathbf{B}}'_w \right) \ell_\kappa \quad (\text{A.12})$$

Second, we examine the asymptotic behavior of the second term on the right side. We first decompose $\phi_t^{(1)}$ into $\bar{\phi}_t^{(1)} + m_h$ such that $m_h = -\frac{h^4}{2} \mu_2 f_u^{(3)}(0) + o(h^4)$; see Section A.1. From this, we obtain that

$$\mathbf{T}_{jj}^{-1} h s_j = \mathbf{T}_{jj}^{-1} h \sum_{t=1}^n \phi_t^{(1)} \mathbf{j}_t = \mathbf{T}_{jj}^{-1} h \sum_{t=1}^n \bar{\phi}_t^{(1)} \mathbf{j}_t + \mathbf{T}_{jj}^{-1} h m_h \sum_{t=1}^n \mathbf{j}_t$$

and examine each term on the right side. First, we focus on the second term on the right side:

$$\mathbf{T}_{jj}^{-1} h m_h \sum_{t=1}^n \mathbf{j}_t = \frac{m_h}{h^2 f_u^{(2)}(0)} \mathbb{E}[\mathbf{j}_t \mathbf{j}'_t]^{-1} \mathbb{E}[\mathbf{j}_t] \{1 + o_{\mathbb{P}}(1)\}$$

by noting that $(nh^3)^{-1} \mathbf{T}_{jj} = f_u^{(2)}(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}'_t] + o_{\mathbb{P}}(1)$ and $n^{-1} \sum_{t=1}^n \mathbf{j}_t = \mathbb{E}[\mathbf{j}_t] + o_{\mathbb{P}}(1)$ using (i) and the ergodic theorem, respectively. Given this, we further note that $\mathbb{E}[\mathbf{j}_t \mathbf{j}'_t]^{-1} \mathbb{E}[\mathbf{j}_t] = \mathbf{e}_1$. Hence, if we let $b_h := m_h / (h^2 f_u^{(2)}(0)) = -\frac{h^2}{2} \mu_2 f_u^{(3)}(0) / f_u^{(2)}(0)$, then it follows that

$$\mathbf{T}_{jj}^{-1} h m_h \sum_{t=1}^n \mathbf{j}_t = b_h \mathbf{e}_1 \{1 + o_{\mathbb{P}}(1)\}. \quad (\text{A.13})$$

Second, we note that

$$\sqrt{nh^3} \mathbf{T}_{jj}^{-1} h \sum_{t=1}^n \bar{\phi}_t^{(1)} \mathbf{j}_t = \left(\frac{1}{nh^3} \mathbf{T}_{jj} \right)^{-1} \left(\frac{1}{\sqrt{nh}} \sum_{t=1}^n \bar{\phi}_t^{(1)} \mathbf{j}_t \right) \rightsquigarrow \ell_\rho \sim N(\mathbf{0}, \Psi^+) \quad (\text{A.14})$$

by noting that $\Psi^+ := \omega_* \mathbb{E}[\mathbf{j}_t \mathbf{j}_t']^{-1}$,

$$\frac{1}{nh^3} \mathbf{T}_{jj} \xrightarrow{\mathbb{P}} f_u^{(2)}(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}_t'] \quad \text{and} \quad \frac{1}{\sqrt{nh}} \sum_{t=1}^n \bar{\phi}_t^{(1)} \mathbf{j}_t \overset{\mathbb{A}}{\sim} N(\mathbf{0}, v^2 f_u(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}_t'])$$

Finally, we combine (A.12) with (A.13) and (A.14) to obtain

$$\sqrt{nh^3}(\mathbf{a}_n - b_h \mathbf{e}_1) = -\sqrt{nh^3} \mathbf{T}_{jj}^{-1} \mathbf{T}_{jx} \mathbf{b}_n + \sqrt{nh^3} \mathbf{T}_{jj}^{-1} h \sum_{t=1}^n \bar{\phi}_t^{(1)} \mathbf{j}_t + o_{\mathbb{P}}(1) \rightsquigarrow \boldsymbol{\ell}_\rho - \mathbf{e}_1 \left(\int \bar{\mathbf{B}}'_w \right) \boldsymbol{\ell}_\kappa,$$

as desired.

(iv) The limits $\boldsymbol{\ell}_\rho$ and $\boldsymbol{\ell}_\kappa$ are driven by $(nh)^{-1/2} \sum_{t=1}^n \bar{\phi}_t^{(1)} \mathbf{j}_t$ and $(n\sqrt{h})^{-1} \sum_{t=1}^n \phi_t^{(1)} \tilde{\mathbf{x}}_t$, respectively. Their predictable asymptotic cross-covariation is proportional to $n^{-3/2} \sum_t \mathbf{j}_t \tilde{\mathbf{x}}_t'$, which converges to zero by Lemma A.7 (vi). Furthermore, both $\boldsymbol{\ell}_\kappa$ and $\boldsymbol{\ell}_\rho$ follow Gaussian probability law, so that $\boldsymbol{\ell}_\rho$ and $\boldsymbol{\ell}_\kappa$ are independent.

(v) Let $\mathbf{A} := [\mathbf{0} : \mathbf{I}_{kq+p}]$, so that $\mathbf{A} \mathbf{e}_1 = \mathbf{0}$. Given (iii),

$$\sqrt{nh^3} \mathbf{A}(\hat{\boldsymbol{\rho}}_n - \boldsymbol{\rho}_\dagger - b_h \mathbf{e}_1) = \sqrt{nh^3} \mathbf{A}(\hat{\boldsymbol{\rho}}_n - \boldsymbol{\rho}_\dagger) \rightsquigarrow \mathbf{A} \boldsymbol{\ell}_\rho - \mathbf{A} \mathbf{e}_1 \left(\int \bar{\mathbf{B}}'_w \right) \boldsymbol{\ell}_\kappa = \mathbf{A} \boldsymbol{\ell}_\rho \sim N(\mathbf{0}, \mathbf{A} \Psi^+ \mathbf{A}')$$

from the fact that $\boldsymbol{\ell}_\rho \sim N(\mathbf{0}, \Psi^+)$. Finally, we note that $\mathbf{A}(\hat{\boldsymbol{\rho}}_n - \boldsymbol{\rho}_\dagger) = ((\hat{\mathbf{v}}_n - \mathbf{v}_\dagger)', (\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_*)')'$, so that $\sqrt{nh^3}((\hat{\mathbf{v}}_n - \mathbf{v}_\dagger)', (\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_*)')' \overset{\mathbb{A}}{\sim} N(\mathbf{0}, \mathbf{A} \Psi^+ \mathbf{A}')$. This completes the proof. $\blacksquare$

Proof of Lemma A.7: (i.a) If we let $\mathbf{H} := P_{\widehat{\mathbf{W}}} \mathbf{K}$, Frisch–Waugh–Lovell's representation implies that

$$P_J = P_{\widehat{\mathbf{W}}} - \mathbf{H}(\mathbf{H}' \mathbf{H})^{-1} \mathbf{H}', \quad \widetilde{\mathbf{X}} = \widetilde{\mathbf{X}} - \mathbf{H} \mathbf{A}_n, \quad \text{where} \quad \mathbf{A}_n := (\mathbf{H}' \mathbf{H})^{-1} \mathbf{H}' \widetilde{\mathbf{X}},$$

because $\text{span}(\mathbf{J}) = \text{span}(\widehat{\mathbf{W}}) \oplus \text{span}(\mathbf{H})$ and $\widehat{\mathbf{W}}' \mathbf{H} = \mathbf{0}$. Given this, the ergodic theorem and the positive definiteness in Assumption 1 (iv.e) imply $n^{-1} \mathbf{H}' \mathbf{H} \xrightarrow{\mathbb{P}} \mathbb{E}[\mathbf{h}_t \mathbf{h}_t']$, which is positive definite. Furthermore, $\mathbf{H}' \widetilde{\mathbf{X}} = O_{\mathbb{P}}(n)$ by Assumption 1 (vi), so that $\mathbf{A}_n = O_{\mathbb{P}}(1)$, which implies that $\sum_{t=1}^n \phi_t^{(2)} \mathbf{h}_t \tilde{\mathbf{x}}_t' = \sum_{t=1}^n \bar{\phi}_t^{(2)} \mathbf{h}_t \tilde{\mathbf{x}}_t' + \sum_{t=1}^n m_h^{(2)} \mathbf{h}_t \tilde{\mathbf{x}}_t' = O_{\mathbb{P}}(nh^3)$, because $\sum_{t=1}^n \bar{\phi}_t^{(2)} \mathbf{h}_t \tilde{\mathbf{x}}_t' = O_{\mathbb{P}}(n\sqrt{h})$ by following the proof of Lemma A.7 (vi), and $\sum_{t=1}^n m_h^{(2)} \mathbf{h}_t \tilde{\mathbf{x}}_t' = O_{\mathbb{P}}(nh^3)$ from $\mathbf{H}' \widetilde{\mathbf{X}} = O_{\mathbb{P}}(n)$. Likewise, $\sum_t \phi_t^{(2)} \mathbf{h}_t \mathbf{h}_t'$ by showing that $\sum_t \bar{\phi}_t^{(2)} \mathbf{h}_t \mathbf{h}_t' = O_{\mathbb{P}}(\sqrt{nh})$ and $\sum_t m_h^{(2)} \mathbf{h}_t \mathbf{h}_t' = O_{\mathbb{P}}(nh^3)$. Therefore,

$$\sum_{t=1}^n \phi_t^{(2)} \mathbf{h}_t \tilde{\mathbf{x}}_t' = O_{\mathbb{P}}(n\sqrt{h}), \quad \text{and} \quad \sum_{t=1}^n \phi_t^{(2)} \mathbf{h}_t \mathbf{h}_t' = O_{\mathbb{P}}(nh^3).$$

because $nh^5 \rightarrow \infty$.

We now note that $\ddot{\mathbf{x}}_t - \tilde{\mathbf{x}}_t = -\mathbf{A}'_n \mathbf{h}_t$, so that $\ddot{\mathbf{x}}_t \ddot{\mathbf{x}}_t = \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' - \mathbf{A}'_n \mathbf{h}_t \tilde{\mathbf{x}}_t' - \tilde{\mathbf{x}}_t \mathbf{h}_t' \mathbf{A}_n + \mathbf{A}'_n \mathbf{h}_t \mathbf{h}_t' \mathbf{A}_n$, further implying that

$$\sum_{t=1}^n \phi_t^{(2)} \ddot{\mathbf{x}}_t \ddot{\mathbf{x}}_t = \sum_{t=1}^n \phi_t^{(2)} \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' - \underbrace{\mathbf{A}'_n \sum_{t=1}^n \phi_t^{(2)} \mathbf{h}_t \tilde{\mathbf{x}}_t'}_{O_{\mathbb{P}}(n\sqrt{h})} - \underbrace{\sum_{t=1}^n \phi_t^{(2)} \tilde{\mathbf{x}}_t \mathbf{h}_t' \mathbf{A}_n}_{O_{\mathbb{P}}(n\sqrt{h})} + \underbrace{\mathbf{A}'_n \sum_{t=1}^n \phi_t^{(2)} \mathbf{h}_t \mathbf{h}_t' \mathbf{A}_n}_{O_{\mathbb{P}}(nh^3)}.$$

Therefore, $\sum_{t=1}^n \phi_t^{(2)} \ddot{\mathbf{x}}_t \ddot{\mathbf{x}}_t - \sum_{t=1}^n \phi_t^{(2)} \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' = o_{\mathbb{P}}(n^2 h^3)$.

(i.b) We note that $\mathbf{H}' \boldsymbol{\iota}_n = \mathbf{0}$, because the intercept is contained in $\widehat{\mathbf{W}}$. Consequently,

$$\frac{1}{n\sqrt{h}} \sum_{t=1}^n \phi_t^{(1)} (\ddot{\mathbf{x}}_t - \tilde{\mathbf{x}}_t) = -\mathbf{A}'_n \frac{1}{n\sqrt{h}} \sum_{t=1}^n \bar{\phi}_t^{(1)} \mathbf{h}_t = O_{\mathbb{P}}(n^{-1/2}) = o_{\mathbb{P}}(1),$$

where the second last equality follows from $\sum_t \bar{\phi}_t^{(1)} \mathbf{h}_t = O_{\mathbb{P}}(\sqrt{nh})$ by applying (A.14). This proves (i.b).

(ii) We split the sum as follows:

$$\sum_{t=1}^n \phi_t^{(2)} \mathbf{x}_t' = m_h^{(2)} \sum_{t=1}^n \mathbf{x}_t' + \sum_{t=1}^n \bar{\phi}_t^{(2)} \mathbf{x}_t'.$$

For the first term, $n^{-3/2} \sum_{t=1}^n \mathbf{x}_t \rightsquigarrow \int \bar{\mathbf{B}}_w(r) dr$ by Lemma A.3 (i), so that it equals $h^3 f_u^{(2)}(0) n^{3/2} \{ \int \bar{\mathbf{B}}_w' + o_{\mathbb{P}}(1) \}$. For the second term, the summands form an MDA because $\mathbf{x}_t$ is $\mathcal{F}_{t-1}$ -measurable, so that

$$\text{var} \left(\sum_{t=1}^n \bar{\phi}_t^{(2)} \mathbf{x}_t' \right) = \sum_{t=1}^n \mathbb{E} [(\bar{\phi}_t^{(2)})^2 \mathbf{x}_t \mathbf{x}_t'] = O(h) \sum_{t=1}^n \mathbb{E} [\mathbf{x}_t \mathbf{x}_t'] = O(hn^2),$$

because $\mathbb{E} \|\mathbf{x}_t\|^2 = O(t)$, so that $\sum_{t=1}^n \bar{\phi}_t^{(2)} \mathbf{x}_t' = O_{\mathbb{P}}(n\sqrt{h})$. Comparing the two terms, the ratio of the second to the first is $O(n\sqrt{h}/(h^3 n^{3/2})) = O((nh^5)^{-1/2}) = o(1)$ as $nh^5 \rightarrow \infty$, so that the first term dominates, and

(ii) follows by dividing $\sum_{t=1}^n \phi_t^{(2)} \mathbf{x}_t'$ by $n^{3/2} h^3$.

(iii) and (iv) Decomposing $\phi_t^{(2)}$ as above and letting $\mathbf{v}_t$ denote $\bar{\mathbf{w}}_t$ in (iii) or $\mathbf{k}_t$ in (iv), the centered part of either statistic is $O_{\mathbb{P}}(n\sqrt{h})$ because

$$\text{var} \left(\sum_{t=1}^n \bar{\phi}_t^{(2)} \mathbf{v}_t \mathbf{x}_t' \right) = O(h) \sum_{t=1}^n \mathbb{E} [\|\mathbf{v}_t\|^2 \|\mathbf{x}_t\|^2] = O(hn^2),$$

where the last equality holds by Assumption 1 (iv.d), the Cauchy–Schwarz inequality, and $\mathbb{E} \|\mathbf{x}_t\|^4 = O(t^2)$.

For (iii), $m_h^{(2)} \sum_t \bar{\mathbf{w}}_t \mathbf{x}_t' = O_{\mathbb{P}}(nh^3) = o_{\mathbb{P}}(n\sqrt{h})$ by Lemma A.2 (ii), proving (iii). For (iv), $\sum_t \mathbf{k}_t \mathbf{x}_t' = O_{\mathbb{P}}(n^{3/2})$ by Lemma A.2 (v). Its systematic part is therefore $O_{\mathbb{P}}(n^{3/2} h^3)$ and dominates $O_{\mathbb{P}}(n\sqrt{h})$ because

$nh^5 \rightarrow \infty$. This proves (iv).

(v) We first note that $\hat{\alpha}_n - \alpha_* = (\hat{\nu}_n - \nu_*) - \mu_*' \iota_p' (\hat{\phi}_n - \phi_*) = s_\alpha' (\hat{\rho}_n - \rho_*)$ and $s_\alpha' e_1 = 1$, so that the desired result follows from Lemma A.6 (iii).

(vi) Because the rows of $P_{\hat{W}} X$ are orthogonal to $\hat{w}_t = (1, \bar{w}_t')'$, $\sum_{t=1}^n \tilde{x}_t = \mathbf{0}$ and $\sum_{t=1}^n \bar{w}_t \tilde{x}_t' = \mathbf{0}$. Therefore,

$$\sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \tilde{x}_t' = \sum_{t=1}^n \begin{bmatrix} \phi_t^{(2)} \tilde{x}_t' \\ \phi_t^{(2)} \bar{w}_t \tilde{x}_t' \\ \phi_t^{(2)} \mathbf{k}_t \tilde{x}_t' \end{bmatrix} = \sum_{t=1}^n \begin{bmatrix} \bar{\phi}_t^{(2)} \tilde{x}_t' \\ \bar{\phi}_t^{(2)} \bar{w}_t \tilde{x}_t' \\ m_h^{(2)} \mathbf{k}_t \tilde{x}_t' + \bar{\phi}_t^{(2)} \mathbf{k}_t \tilde{x}_t' \end{bmatrix} = \begin{bmatrix} O_{\mathbb{P}}(n\sqrt{h}) \\ O_{\mathbb{P}}(n\sqrt{h}) \\ O_{\mathbb{P}}(nh^3) + O_{\mathbb{P}}(n\sqrt{h}) \end{bmatrix}.$$

Here, we use the fact that $\sum_t \mathbf{k}_t \tilde{x}_t' = O_{\mathbb{P}}(n)$ that holds by applying Lemma A.2 (v). Therefore, $(n^{3/2}h^3)^{-1} \sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \tilde{x}_t' = o_{\mathbb{P}}(1)$. This completes the proof. ■

A.3 Proofs

In this section, we prove the main claims of this study.

Proof of Theorem 1: The autoregressive parameter ϕ_* is identical to the coefficient of $\mathbf{k}_t$ in the reparameterized model, viz., ϕ_* , and the estimators for them is identical, which is given as $\hat{\phi}_n$ in $\hat{\rho}_n$. Lemma A.6 (v) directly derives the joint limit distribution of $\hat{\rho}_n$, so that $\sqrt{nh^3}(\hat{\phi}_n - \phi_*) \overset{A}{\rightsquigarrow} N(\mathbf{0}, \Psi_{\phi\phi})$, where $\Psi_{\phi\phi}$ is the last $p \times p$ block of Ψ . By (16) and the definition of $\mathbf{h}_t$, $\Psi_{\phi\phi} = \omega_* \mathbb{E}[\mathbf{h}_t \mathbf{h}_t']^{-1} = \Pi$. This completes the proof. ■

Proof of Theorem 2: (i) By the reparameterization given in (13), we note that $(\hat{\gamma}_n - \gamma_*) + \beta_*' \iota_p' (\hat{\phi}_n - \phi_*) = \hat{\kappa}_n - \beta_*$, and Lemma A.6 (ii) already provides the desired weak limit.

(ii) If we note that $(\hat{\gamma}_n - \gamma_*) + \beta_*' \iota_p' (\hat{\phi}_n - \phi_*) = \hat{\kappa}_n - \beta_*$,

$$\sqrt{nh^3}(\hat{\gamma}_n - \gamma_*) = \sqrt{nh^3}(\hat{\kappa}_n - \beta_*) - \beta_*' \iota_p' \sqrt{nh^3}(\hat{\phi}_n - \phi_*) = -\beta_*' \iota_p' \sqrt{nh^3}(\hat{\phi}_n - \phi_*) + o_{\mathbb{P}}(1),$$

where the second equality follows from the fact that $(\hat{\kappa}_n - \beta_*) = O_{\mathbb{P}}((n^2h^3)^{-1/2})$. Furthermore, Theorem 1 already shows that $\sqrt{nh^3}(\hat{\phi}_n - \phi_*) \overset{A}{\rightsquigarrow} N(\mathbf{0}, \Pi)$, so that $\sqrt{nh^3}(\hat{\gamma}_n - \gamma_*) \overset{A}{\rightsquigarrow} N(\mathbf{0}, \sigma_\phi^2 \beta_* \beta_*')$ with $\sigma_\phi^2 = \iota_p' \Pi \iota_p$ as desired. ■

Proof of Theorem 3: From (21) and $\sum_{j=1}^p (\hat{\phi}_{j,n} - \phi_{j*}) = \boldsymbol{\iota}'_p (\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_*)$, it follows that

$$\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}_* = \left[1 - \sum_{j=1}^p \hat{\phi}_{j,n} \right]^{-1} \left[(\hat{\gamma}_n - \gamma_*) + \boldsymbol{\beta}_* \boldsymbol{\iota}'_p (\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_*) \right] = \left[1 - \sum_{j=1}^p \hat{\phi}_{j,n} \right]^{-1} (\hat{\boldsymbol{\kappa}}_n - \boldsymbol{\beta}_*).$$

Given this, $\sum_{j=1}^p \hat{\phi}_{j,n} \xrightarrow{\mathbb{P}} \sum_{j=1}^p \phi_{j*}$ and the continuous mapping theorem applied to Theorem 2 (i) lead to the desired limit distribution. ■

Proof of Theorem 4: Since $\tilde{\boldsymbol{\varphi}}_n := -\mathbf{R}_1 \hat{\boldsymbol{\phi}}_n$, $\sqrt{nh^3}(\tilde{\boldsymbol{\varphi}}_n - \boldsymbol{\varphi}_*) = -\mathbf{R}_1 \sqrt{nh^3}(\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_*) \stackrel{\Delta}{\sim} N(\mathbf{0}, \mathbf{R}_1 \boldsymbol{\Pi} \mathbf{R}'_1)$ by Theorem 1. This completes the proof. ■

Proof of Theorem 5: (i) By the reparameterization, we obtain that

$$\hat{\boldsymbol{\delta}}_n - \boldsymbol{\delta}_* = (\hat{\mathbf{v}}_n - \mathbf{v}_\dagger) - \boldsymbol{\Xi}_* (\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_*) = [\mathbf{I}_{kq} \vdots -\boldsymbol{\Xi}_*] ((\hat{\mathbf{v}}_n - \boldsymbol{\delta}_\dagger)', (\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_*)')'.$$

By Lemma A.6 (v), it follows that $\sqrt{nh^3}((\hat{\mathbf{v}}_n - \mathbf{v}_\dagger)', (\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_*)')' \stackrel{\Delta}{\sim} N(\mathbf{0}, \boldsymbol{\Psi})$. Therefore, $\sqrt{nh^3}(\hat{\boldsymbol{\delta}}_n - \boldsymbol{\delta}_*) \stackrel{\Delta}{\sim} N(\mathbf{0}, \boldsymbol{\Omega}_\delta)$ with $\boldsymbol{\Omega}_\delta = [\mathbf{I}_{kq} \vdots -\boldsymbol{\Xi}_*] \boldsymbol{\Psi} [\mathbf{I}_{kq} \vdots -\boldsymbol{\Xi}_*]'$.

(ii) Using (20) and the fact that $\tilde{\boldsymbol{\psi}}_n = \hat{\boldsymbol{\delta}}_n + \mathbf{S} \hat{\boldsymbol{\gamma}}_n$, $\boldsymbol{\psi}_* = \boldsymbol{\delta}_* + \mathbf{S} \boldsymbol{\gamma}_*$, we obtain that

$$\sqrt{nh^3}(\tilde{\boldsymbol{\psi}}_n - \boldsymbol{\psi}_*) = [\mathbf{I}_{kq} \vdots -\mathbf{C}] \sqrt{nh^3} \begin{pmatrix} \hat{\mathbf{v}}_n - \boldsymbol{\delta}_\dagger \\ \hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_* \end{pmatrix} + O_{\mathbb{P}}(n^{-1/2}).$$

Here, the remainder term is $O_{\mathbb{P}}(n^{-1/2})$, because $\mathbf{S}(\hat{\boldsymbol{\kappa}}_n - \boldsymbol{\beta}_*) = O_{\mathbb{P}}((n\sqrt{h^3})^{-1})$. Therefore, $\sqrt{nh^3}(\tilde{\boldsymbol{\psi}}_n - \boldsymbol{\psi}_*) \stackrel{\Delta}{\sim} N(\mathbf{0}, \boldsymbol{\Omega}_\psi)$ by Lemma A.6 (v). This completes the proof. ■

Proof of Corollary 1. (i) By Theorem 3 (ii), we obtain that

$$\left(1 - \sum_{j=1}^p \phi_{j*} \right) n\sqrt{h^3} (\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}_*) \stackrel{\Delta}{\sim} N \left(\mathbf{0}, \frac{v^2 f_u(0)}{[f_u^{(2)}(0)]^2} \left[\int \tilde{\boldsymbol{\mathcal{B}}}_w \tilde{\boldsymbol{\mathcal{B}}}_w' \right]^{-1} \right)$$

conditional on $\overline{\boldsymbol{\mathcal{B}}}_w(\cdot)$. The weight matrix of $\mathcal{W}_n(\boldsymbol{\beta})$ weakly converges to the conditional covariance by Lemma A.4 (iv):

$$\hat{\omega}_n \left(\frac{1}{n^2} \sum_{t=1}^n \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \right)^{-1} \rightsquigarrow \omega_* \left[\int \tilde{\boldsymbol{\mathcal{B}}}_w \tilde{\boldsymbol{\mathcal{B}}}_w' \right]^{-1},$$

because $\hat{f}_u(0) \xrightarrow{\mathbb{P}} f_u(0)$ and $\hat{f}_u^{(2)}(0) \xrightarrow{\mathbb{P}} f_u^{(2)}(0)$. Furthermore, $\sum_{j=1}^p \hat{\phi}_{j,n} \xrightarrow{\mathbb{P}} \sum_{j=1}^p \phi_{j*}$ by Theorem 1 (i). Therefore, $\mathcal{W}_n(\boldsymbol{\beta})$ asymptotically follows $\mathcal{X}_r^2$ conditional on $\boldsymbol{\mathcal{B}}_w(\cdot)$. Furthermore, the conditional weak limit

does not depend on $\bar{\mathcal{B}}_w(\cdot)$, so that it asymptotically follows $\mathcal{X}_r^2$ unconditionally.

(ii) Under $\mathbb{H}_1^{(1)}$, $\mathcal{Q}\beta_* - \mathbf{q} \neq \mathbf{0}$ and $\mathcal{W}_n(\beta)$ diverges at the rate n^2h^3 , so that $\mathbb{P}(\mathcal{W}_n(\beta) \geq s_n^{(1)}) \xrightarrow{\mathbb{P}} 1$ for any $s_n^{(1)} = o(n^2h^3)$. This completes the proof. ■

Proof of Corollary 3. (i) We prove both (i.a) and (i.b). If $\mathcal{Q}\beta_* = \mathbf{0}$, then (22) reduces to $\mathcal{Q}(\hat{\gamma}_n - \gamma_*) = \mathcal{Q}(\hat{\kappa}_n - \beta_*)$, so that we obtain the following by Theorem 2 (i),

$$n\sqrt{h^3} \mathcal{Q}(\hat{\gamma}_n - \gamma_*) \stackrel{\mathbb{A}}{\sim} N\left(\mathbf{0}, \mathcal{Q} \frac{v^2 f_u(0)}{[f_u^{(2)}(0)]^2} \left[\int \tilde{\mathcal{B}}_w \tilde{\mathcal{B}}_w' \right]^{-1} \mathcal{Q}'\right)$$

conditional on $\bar{\mathcal{B}}_w(\cdot)$. The weight matrix of $\mathcal{W}_n^{(0)}(\gamma)$ converges to the conditional covariance by Lemma A.4

(iv) together with the consistency of $\hat{\omega}_n$ for ω_* . Therefore, $\mathcal{W}_n^{(0)}(\gamma) \stackrel{\mathbb{A}}{\sim} \mathcal{X}_r^2$ under $\mathbb{H}_1^{(3)}$. Under $\mathbb{H}_1^{(3)}$, $\mathcal{Q}\gamma_* - \mathbf{q} \neq \mathbf{0}$ and the Wald test diverges at the rate of n^2h^3 , so that $\mathbb{P}(\mathcal{W}_n^{(0)}(\gamma) \geq s_n) \xrightarrow{\mathbb{P}} 1$ for any $s_n = o(n^2h^3)$.

(ii) Without loss of generality, we let $\mathcal{Q} = \mathbf{q}'_1$. If $\mathcal{Q} = \mathbf{q}'_1$ such that $\mathbf{q}'_1\beta_* \neq 0$, then

$$\sqrt{nh^3} \mathbf{q}'_1(\hat{\gamma}_n - \gamma_*) = \sqrt{nh^3} \mathbf{q}'_1(\hat{\kappa}_n - \beta_*) - (\mathbf{q}'_1\beta_*) \boldsymbol{\iota}'_p \sqrt{nh^3} (\hat{\phi}_n - \phi_*)$$

by (22). The first term on the right side is $n^{-1/2} \cdot n\sqrt{h^3} \mathbf{q}'_1(\hat{\kappa}_n - \beta_*) = O_{\mathbb{P}}(n^{-1/2}) = o_{\mathbb{P}}(1)$ by Theorem 2 (i), and $\boldsymbol{\iota}'_p \sqrt{nh^3} (\hat{\phi}_n - \phi_*) \stackrel{\mathbb{A}}{\sim} N(0, \sigma_\phi^2)$ by Theorem 1. Therefore, it follows that

$$\sqrt{nh^3} \mathbf{q}'_1(\hat{\gamma}_n - \gamma_*) \stackrel{\mathbb{A}}{\sim} N(0, \sigma_\phi^2 (\mathbf{q}'_1\beta_*)^2).$$

Since $\hat{\sigma}_\phi^2 \xrightarrow{\mathbb{P}} \sigma_\phi^2$ and $\hat{\beta}_n \xrightarrow{\mathbb{P}} \beta_*$, Slutsky's theorem delivers $\mathcal{W}_n^{(1)}(\gamma) \stackrel{\mathbb{A}}{\sim} \mathcal{X}_1^2$ under $\mathbb{H}_0^{(3)} : \mathbf{q}'_1\gamma_* = q_1$. Under $\mathbb{H}_1^{(3)}$, $\mathbf{q}'_1\gamma_* - q_1 \neq 0$ and the Wald test diverges at the rate of nh^3 , so that $\mathbb{P}(\mathcal{W}_n^{(1)}(\gamma) \geq s_n) \xrightarrow{\mathbb{P}} 1$ for any $s_n = o(nh^3)$. ■

A.4 Supplementary Simulations

In this section, we provide supplementary simulation results.

A.4.1 Simulation Using MARDL Model

We assume the same DGPs and model conditions as in Section 3.1, and report the simulation results for the estimators using the MARDL model. First, Figure A.1 shows the empirical densities of the estimators obtained by simulation under DGPs 1, 2, and 3. Each estimator is standardized so that it follows the standard

normal distribution:

$$\frac{\sqrt{nh^3}(\hat{\phi}_n - \phi_*)}{\hat{\Pi}_n^{1/2}}, \quad \frac{\sqrt{nh^3}(\hat{\gamma}_n - \gamma_*)}{\hat{\sigma}_{\phi_n}|\hat{\beta}_n|}, \quad \text{and} \quad \frac{n\sqrt{h^3}(\hat{\beta}_n - \beta_*)}{|1 - \hat{\phi}_n|^{-1}\hat{\omega}_n^{1/2}[n^{-2}\sum_{t=1}^n\tilde{x}_t^2]^{-1/2}}.$$

Figure A.1 shows the empirical densities of these three statistics for $n = 200, 400, 600, 800, 1000, 2000$, and 3000 and we observe that they approach the standard normal density function as the sample size increases, corroborating Theorems 1, 2 (ii), and 3.

Second, Figure A.2 presents the QQ plots between the three statistics and the standard normal random variable for $n = 200, 1000$, and 3000 . We also observe that the QQ plot approaches the 45-degree line as the sample size grows. This corroborates the asymptotic theory on the parameter estimators in Theorems 1, 2 (ii), and 3.

Finally, we conduct 1000 independent experiments for different sample sizes $n = 200, 400, 600, 800, 1000, 2000$ and 3000 , and report the sample average of the estimated coefficients and their MSEs in Table A.1. For all three DGPs, all MSEs converge to zero, implying that the estimated parameters are consistent for the DGP parameters. Regardless of the error distribution, the modal estimators exhibit slight bias for small n , but significant improvements are attained as the sample size increases. This corroborates the consistency of the estimators. In particular, the MSE of $\hat{\beta}_n$ decreases more rapidly than the other estimators because the long-run parameter estimator is super-consistent.

A.4.2 Simulation Using MARDL-EC

We assume the same DGPs and model conditions as in Section 3.2, and report the simulation results for the estimators using the MARDL-EC representation. First, Figure A.3 shows the empirical density functions of the standardized EC parameter estimators under DGPs 1, 2, and 3. The standardized EC parameter estimators are obtained by using the sample analogs of the asymptotic covariance matrices given in Theorems 4 and 5. As the sample size increases, the empirical density functions approach the standard normal density function for both $\tilde{\varphi}_n$ and $\tilde{\psi}_n$. This feature corroborates Theorems 4 and 5.

Second, Figure A.4 presents the QQ plots between the standardized statistics and the standard normal random variable under DGPs 1, 2, and 3. As the sample size increases, the QQ plot approaches the 45-degree line. This also corroborates Theorems 4 and 5.

Finally, we examine the finite-sample bias and MSEs of the MARDL-EC parameter estimators in parallel to Table A.1. We report the simulation results in Table A.2. For all the cases we consider, we observe consistent decreases in MSE as the sample size grows. This indicates that the estimators are consistent for

the MARDL-EC parameters.

A.5 Supplementary Empirics

This section provides supplementary empirical information.

A.5.1 Empirical Data Source

We obtain global COVID-19 data from Our World in Data, primarily sourced from the World Health Organization (WHO).¹ This dataset provides COVID-19 confirmed cases, deaths, and recoveries from various countries since January 3, 2020. The confirmed cases and deaths are collected from the WHO's Coronavirus Dashboard; this dataset is updated daily.

We also obtain other variables related to COVID-19 from Our World in Data² on location, population, GDP per capita (gross domestic product at purchasing power parity, constant 2011 international dollars, most recent year available), population density (number of people divided by land area, measured in square km, most recent year available), median age (median age of the population, UN projection for 2020), aged 65 and older (share of the population that is 65 years and older, most recent year available), aged 70 and older (share of the population that is 70 years and older in 2015), female smokers (share of women who smoke, most recent year available), male smokers (share of men who smoke, most recent year available), hospital beds per thousand (hospital beds per 1000 people, most recent year available since 2010), life expectancy (life expectancy at birth in 2019), and human development index (a composite index measuring average achievement in three basic dimensions of human development in 2019).

Furthermore, the Oxford COVID-19 Government Response Tracker (OxCGRT) systematically gathers information regarding the timing and nature of pandemic response measures enacted by governments. OxCGRT meticulously compiled data on diverse policy responses throughout the years 2020, 2021, and 2022, reflecting the breadth of government actions.³ The data are aggregated to form a series of policy indices, with the Government Response Index (GRI) assessing the government's level of intervention across several policy domains such as

C - Containment and closure policies;

E - Economic policies;

H - Health system policies; and

¹The data are available at <https://ourworldindata.org/covid-vaccinations>

²Data are available at <https://github.com/owid/covid-19-data/tree/master>

³Data are available at <https://www.bsg.ox.ac.uk/research/covid-19-government-response-tracker>

V - Vaccination policies.

These indices gauge the extent to which governments adopted various policy approaches and the degree to which they were implemented. Higher values indicate stricter measures, reflecting the comprehensiveness and stringency of government interventions across the specified policy domains. Note that this index does not provide an evaluation of the effectiveness of the government's policies.

A.5.2 Descriptive Statistics

We provide the descriptive statistics related to COVID-19 for Belgium, Denmark, Germany, and Sweden in Table [A.3](#).

Denmark and Sweden are two European countries comparable in terms of economic strength and the variables related to COVID-19. Their per capita GDP levels are almost identical: \$46,683 for Denmark and \$46,949 for Sweden. Both countries belong to the Scandinavian region and are situated adjacent to the Scandinavian Peninsula. They share relatively low population densities, with Denmark at 137/km² and Sweden at 25/km², which are lower than Belgium and Germany. The two countries also exhibit significant similarities in variables related to COVID-19, such as the proportion of elderly population, smoking rates, and hospital bed capacity. Meanwhile, they differ in population size: 5,882,259 for Denmark against 10,549,349 for Sweden, but the series we analyze are measured per million people, so that this difference does not enter the comparison directly.

By contrast, Belgium and Germany belong to a group of European countries with more concentrated populations and are geographically adjacent to each other. Belgium has a population density of 376/km², while Germany's population density is 237/km². Belgium's population is 11,655,923, while Germany's population is 83,369,840. The economic development levels of the two countries also show similarities, with per capita GDP at \$42,659 for Belgium and \$45,229 for Germany. They also demonstrate noticeable similarities in variables related to COVID-19.

DGP	Para.	Stat. \ n	200	400	600	800	1000	2000	3000
1	α_*	Est.	-0.0087	-0.0018	-0.0058	0.0062	0.0038	0.0009	-0.0012
		Std. Err.	(0.1305)	(0.0983)	(0.0935)	(0.0794)	(0.0744)	(0.0590)	(0.0548)
		MSE	0.0171	0.0097	0.0088	0.0063	0.0055	0.0035	0.0030
	β_*	Est.	2.0007	2.0034	2.0022	1.9987	2.0002	1.9999	1.9999
		Std. Err.	(0.0969)	(0.0496)	(0.0345)	(0.0296)	(0.0251)	(0.0137)	(0.0093)
		MSE	0.0094	0.0025	0.0012	0.0009	0.0006	0.0002	0.0001
	γ_*	Est.	1.0097	1.0026	1.0122	1.0110	1.0043	1.0045	0.9999
		Std. Err.	(0.2015)	(0.1480)	(0.1267)	(0.1204)	(0.1069)	(0.0855)	(0.0788)
		MSE	0.0406	0.0219	0.0162	0.0146	0.0114	0.0073	0.0062
	ϕ_*	Est.	0.4949	0.4993	0.4943	0.4941	0.4979	0.4977	0.5000
		Std. Err.	(0.1004)	(0.0744)	(0.0635)	(0.0604)	(0.0536)	(0.0429)	(0.0393)
		MSE	0.0101	0.0055	0.0041	0.0037	0.0029	0.0018	0.0015
2	α_*	Est.	0.0323	0.0275	0.0224	0.0206	0.0186	0.0155	0.0141
		Std. Err.	(0.0378)	(0.0301)	(0.0263)	(0.0229)	(0.0226)	(0.0170)	(0.0150)
		MSE	0.0025	0.0017	0.0012	0.0009	0.0009	0.0005	0.0004
	β_*	Est.	2.0000	2.0009	2.0006	1.9996	2.0002	2.0000	2.0000
		Std. Err.	(0.0252)	(0.0138)	(0.0103)	(0.0075)	(0.0065)	(0.0036)	(0.0024)
		MSE	0.0006	0.0002	0.0001	0.0001	0.0000	0.0000	0.0000
	γ_*	Est.	1.0022	1.0020	0.9998	1.0015	1.0003	1.0013	1.0010
		Std. Err.	(0.0706)	(0.0564)	(0.0464)	(0.0438)	(0.0400)	(0.0322)	(0.0290)
		MSE	0.0050	0.0032	0.0021	0.0019	0.0016	0.0010	0.0008
	ϕ_*	Est.	0.4988	0.4992	0.5002	0.4991	0.4999	0.4993	0.4995
		Std. Err.	(0.0358)	(0.0285)	(0.0232)	(0.0221)	(0.0201)	(0.0161)	(0.0145)
		MSE	0.0013	0.0008	0.0005	0.0005	0.0004	0.0003	0.0002
3	α_*	Est.	0.3487	0.2586	0.2259	0.2070	0.1979	0.1706	0.1412
		Std. Err.	(0.4358)	(0.2953)	(0.2626)	(0.2313)	(0.2109)	(0.1641)	(0.1482)
		MSE	0.3113	0.1540	0.1199	0.0963	0.0836	0.0560	0.0419
	β_*	Est.	1.9995	2.0008	1.9994	2.0004	1.9987	2.0004	2.0007
		Std. Err.	(0.2527)	(0.1254)	(0.0979)	(0.0716)	(0.0558)	(0.0308)	(0.0227)
		MSE	0.0638	0.0157	0.0096	0.0051	0.0031	0.0009	0.0005
	γ_*	Est.	1.0146	1.0046	1.0013	0.9984	1.0033	1.0022	0.9966
		Std. Err.	(0.2264)	(0.1563)	(0.1258)	(0.1230)	(0.1064)	(0.0825)	(0.0781)
		MSE	0.0514	0.0244	0.0158	0.0151	0.0113	0.0068	0.0061
	ϕ_*	Est.	0.4928	0.4979	0.4989	0.5009	0.4980	0.4990	0.5019
		Std. Err.	(0.0945)	(0.0724)	(0.0609)	(0.0590)	(0.0520)	(0.0409)	(0.0386)
		MSE	0.0090	0.0052	0.0037	0.0035	0.0027	0.0017	0.0015

Table A.1: SIMULATION RESULTS FOR THE MARDL ESTIMATOR. Notes: (i) Number of replications: 1000. Empirical standard deviations are in parentheses and MSE denotes the Monte Carlo mean squared error. (ii) The DGP and true parameter values are as in Table 1. (iii) DGP 1: $u_t \sim_{\text{iid}} N(0, 0.25)$; DGP 2: $u_t \sim_{\text{iid}} \text{Beta}(2, 6) - 1/6$; DGP 3: $u_t \sim_{\text{iid}} \Gamma(3, 1) - 2$.

DGP	Para.	Stat. \ n	200	400	600	800	1000	2000	3000
1	α_*	Est.	-0.0040	0.0005	-0.0015	-0.0015	0.0001	0.0015	-0.0035
		Std. Err.	(0.1308)	(0.0997)	(0.0948)	(0.0833)	(0.0776)	(0.0609)	(0.0548)
		MSE	0.0171	0.0099	0.0090	0.0069	0.0060	0.0037	0.0030
	β_*	Est.	2.0014	1.9994	1.9991	2.0000	2.0008	1.9997	2.0001
		Std. Err.	(0.0946)	(0.0493)	(0.0370)	(0.0286)	(0.0235)	(0.0138)	(0.0097)
		MSE	0.0089	0.0024	0.0014	0.0008	0.0006	0.0002	0.0001
	ζ_*	Est.	-0.5123	-0.5028	-0.5019	-0.5036	-0.5017	-0.4981	-0.5017
		Std. Err.	(0.0921)	(0.0772)	(0.0662)	(0.0563)	(0.0570)	(0.0425)	(0.0383)
		MSE	0.0086	0.0060	0.0044	0.0032	0.0033	0.0018	0.0015
	φ_*	Est.	0.0026	-0.0001	-0.0041	-0.0014	-0.0037	-0.0010	-0.0000
		Std. Err.	(0.1087)	(0.0868)	(0.0747)	(0.0671)	(0.0660)	(0.0504)	(0.0476)
		MSE	0.0118	0.0075	0.0056	0.0045	0.0044	0.0025	0.0023
	ψ_*	Est.	0.4959	0.4972	0.5035	0.4968	0.5011	0.5005	0.5000
		Std. Err.	(0.2783)	(0.2094)	(0.1853)	(0.1653)	(0.1615)	(0.1227)	(0.1098)
		MSE	0.0774	0.0438	0.0343	0.0273	0.0260	0.0150	0.0120
2	α_*	Est.	0.0328	0.0269	0.0235	0.0209	0.0189	0.0152	0.0146
		Std. Err.	(0.0382)	(0.0316)	(0.0267)	(0.0236)	(0.0223)	(0.0175)	(0.0150)
		MSE	0.0025	0.0017	0.0013	0.0010	0.0009	0.0005	0.0004
	β_*	Est.	1.9986	2.0001	1.9993	2.0001	2.0003	2.0000	2.0000
		Std. Err.	(0.0264)	(0.0146)	(0.0106)	(0.0080)	(0.0067)	(0.0039)	(0.0026)
		MSE	0.0007	0.0002	0.0001	0.0001	0.0000	0.0000	0.0000
	ζ_*	Est.	-0.5036	-0.4999	-0.4985	-0.4999	-0.5007	-0.5000	-0.5000
		Std. Err.	(0.0551)	(0.0436)	(0.0369)	(0.0326)	(0.0308)	(0.0245)	(0.0221)
		MSE	0.0030	0.0019	0.0014	0.0011	0.0009	0.0006	0.0005
	φ_*	Est.	-0.0023	0.0017	0.0012	0.0003	-0.0022	-0.0004	0.0009
		Std. Err.	(0.0820)	(0.0641)	(0.0552)	(0.0479)	(0.0458)	(0.0362)	(0.0326)
		MSE	0.0067	0.0041	0.0030	0.0023	0.0021	0.0013	0.0011
	ψ_*	Est.	0.4983	0.5012	0.5001	0.4986	0.5015	0.5012	0.4994
		Std. Err.	(0.0797)	(0.0591)	(0.0523)	(0.0475)	(0.0444)	(0.0344)	(0.0304)
		MSE	0.0063	0.0035	0.0027	0.0023	0.0020	0.0012	0.0009
3	α_*	Est.	0.3342	0.2672	0.2295	0.2278	0.1998	0.1643	0.1494
		Std. Err.	(0.4216)	(0.3350)	(0.2610)	(0.2444)	(0.2209)	(0.1775)	(0.1552)
		MSE	0.2893	0.1835	0.1207	0.1116	0.0886	0.0585	0.0464
	β_*	Est.	2.0028	2.0040	2.0003	1.9983	2.0006	1.9984	1.9994
		Std. Err.	(0.2517)	(0.1353)	(0.0908)	(0.0686)	(0.0592)	(0.0308)	(0.0238)
		MSE	0.0633	0.0183	0.0082	0.0047	0.0035	0.0009	0.0006
	ζ_*	Est.	-0.5039	-0.4994	-0.5040	-0.5022	-0.5013	-0.5017	-0.4981
		Std. Err.	(0.1103)	(0.0850)	(0.0726)	(0.0634)	(0.0588)	(0.0468)	(0.0417)
		MSE	0.0122	0.0072	0.0053	0.0040	0.0035	0.0022	0.0017
	φ_*	Est.	-0.0001	-0.0018	0.0032	0.0018	0.0020	0.0014	-0.0004
		Std. Err.	(0.1075)	(0.0854)	(0.0697)	(0.0660)	(0.0599)	(0.0465)	(0.0400)
		MSE	0.0115	0.0073	0.0049	0.0044	0.0036	0.0022	0.0016
	ψ_*	Est.	0.5057	0.5167	0.4729	0.5096	0.4938	0.4993	0.5009
		Std. Err.	(0.6970)	(0.5458)	(0.4642)	(0.4276)	(0.3648)	(0.3005)	(0.2618)
		MSE	0.4854	0.2979	0.2160	0.1828	0.1330	0.0902	0.0685

Table A.2: SIMULATION RESULTS FOR THE MARDL–EC ESTIMATOR. Notes: (i) Number of replications: 1000. Empirical standard deviations are in parentheses and MSE denotes the Monte Carlo mean squared error. (ii) The DGP and true values are $(\alpha_{3*}, \beta_{2*}, \zeta_*, \varphi_*, \psi_*) = (0, 2, -0.5, 0, 0.5)$, as in Table 3. (iii) The three error distributions are as in Table 3.

	Denmark	Sweden	Belgium	Germany
Population	5,882,259	10,549,349	11,655,923	83,369,840
GDP per capita	46,683	46,949	42,659	45,229
Population density	137	25	376	237
Median age	42	41	42	47
Aged 65 older	20	20	19	21
Aged 70 older	12	13	13	16
Female smokers	19	19	25	28
Male smokers	19	19	31	33
Hospital beds per thousand	3	2	6	8
Life expectancy	81	83	82	81
Human development index	0.94	0.95	0.93	0.95

Table A.3: STATISTICS OF THE VARIABLES RELATED TO COVID-19 FOR THE FOUR EUROPEAN COUNTRIES. The data are sourced from Our World in Data (<https://github.com/owid/covid-19-data/tree/master>).

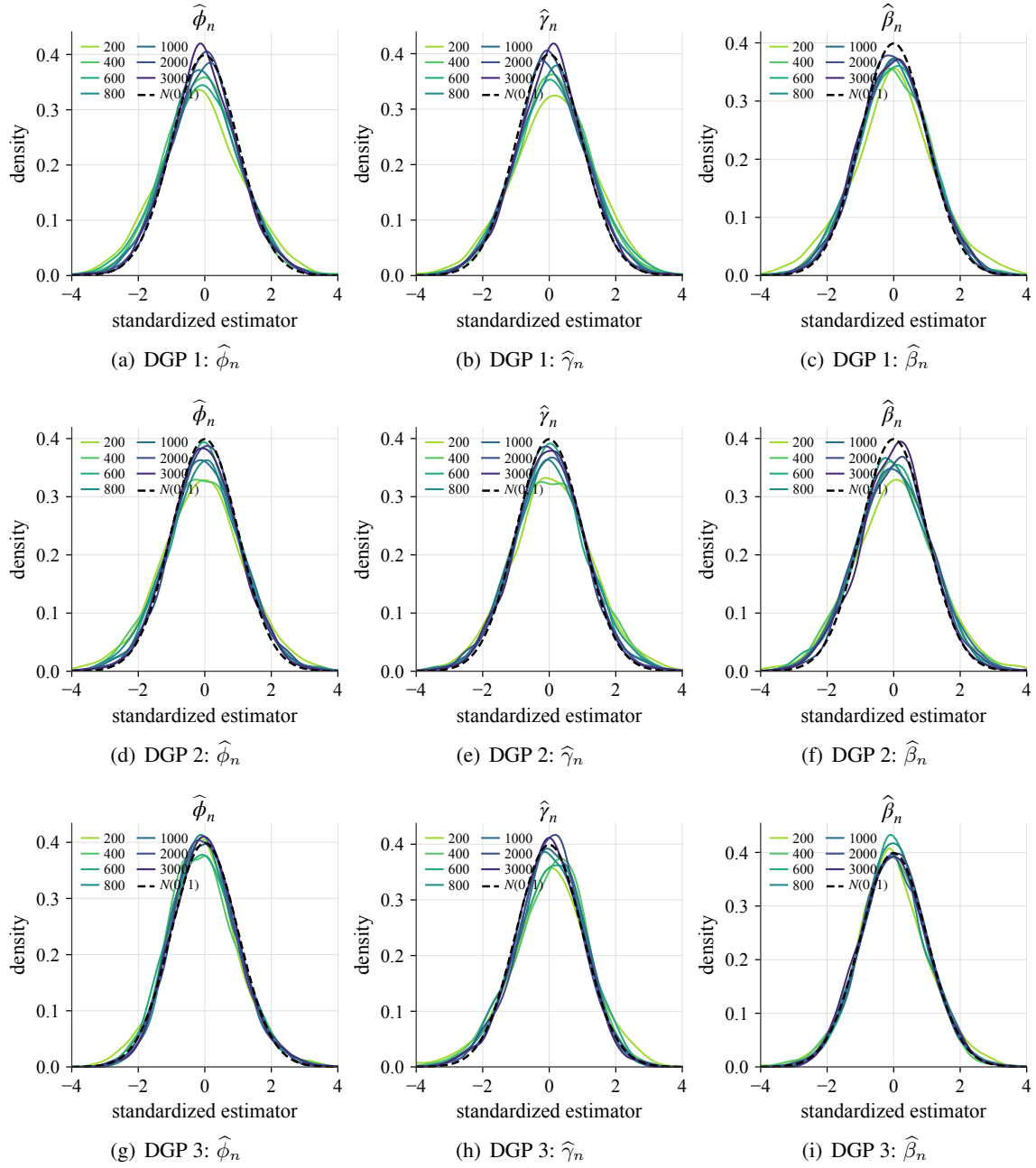


Figure A.1: SAMPLING DENSITIES OF THE MARDL ESTIMATORS UNDER DGPs 1–3. The figures display the probability densities of $\hat{\phi}_n$, $\hat{\gamma}_n$ and $\hat{\beta}_n$. The seven sample sizes range from $n = 200$ to $n = 3000$, with $R = 1500$ iterations at each sample size. DGP 1: $u_t \sim_{\text{iid}} N(0, 0.25)$; DGP 2: $u_t \sim_{\text{iid}} \text{Beta}(2, 6) - 1/6$; DGP 3: $u_t \sim_{\text{iid}} \Gamma(3, 1) - 2$.

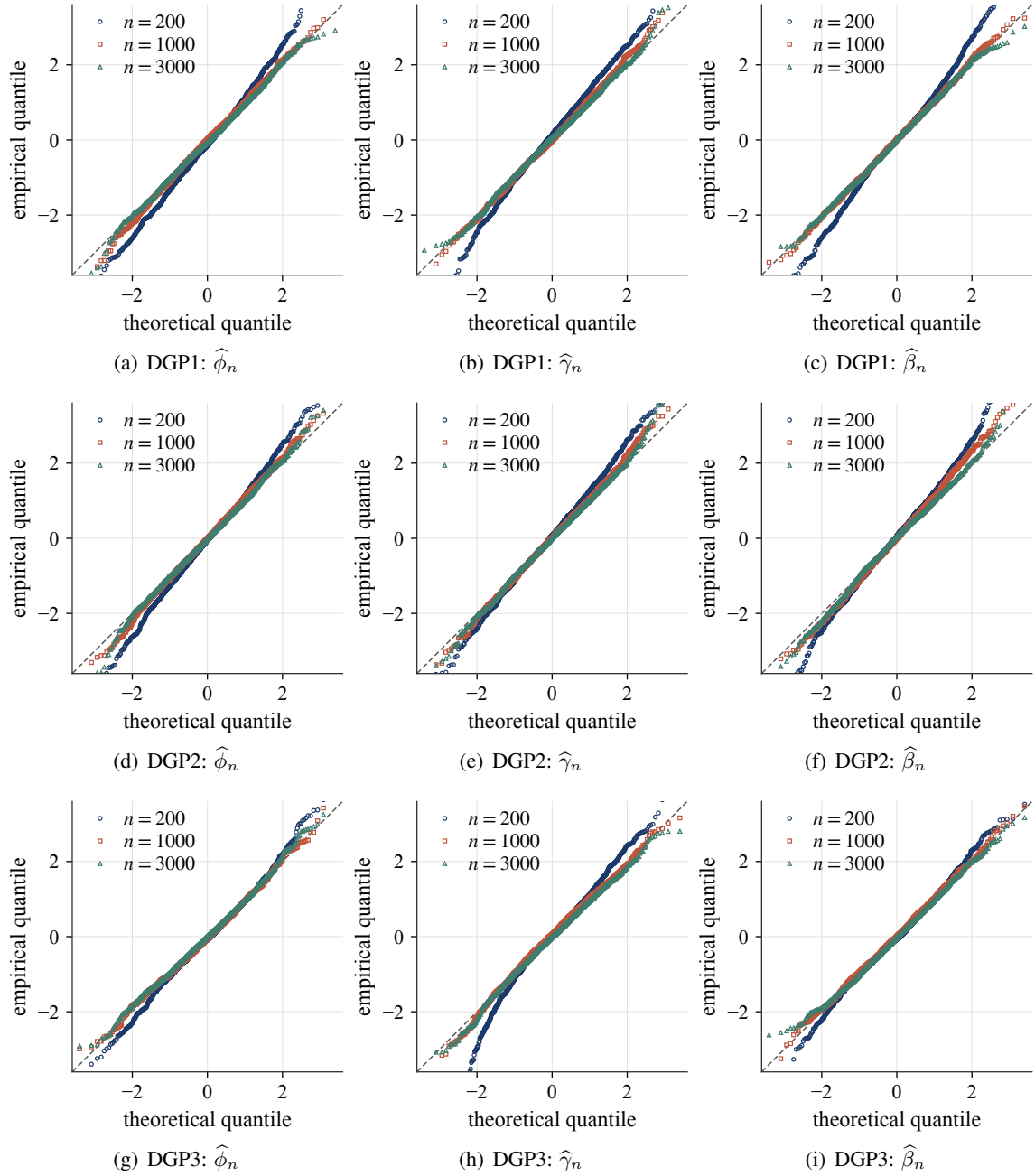


Figure A.2: QQ PLOTS OF THE STANDARDIZED MARDL ESTIMATORS UNDER DGPs 1–3. Notes: (i) $R = 1500$ iterations at each sample size. (ii) Each estimator is centered at its probability limit and divided by its estimated asymptotic standard error. (iii) The horizontal axis gives standard-normal quantiles and the vertical axis the quantiles of the simulated standardized estimator; the dashed line is the 45-degree line. (iv) The plots approach the 45-degree line as the sample size increases, with the remaining discrepancies concentrated mainly in the tails. (v) DGP 1: $u_t \sim_{\text{iid}} N(0, 0.25)$; DGP 2: $u_t \sim_{\text{iid}} \text{Beta}(2, 6) - 1/6$; DGP 3: $u_t \sim_{\text{iid}} \Gamma(3, 1) - 2$.

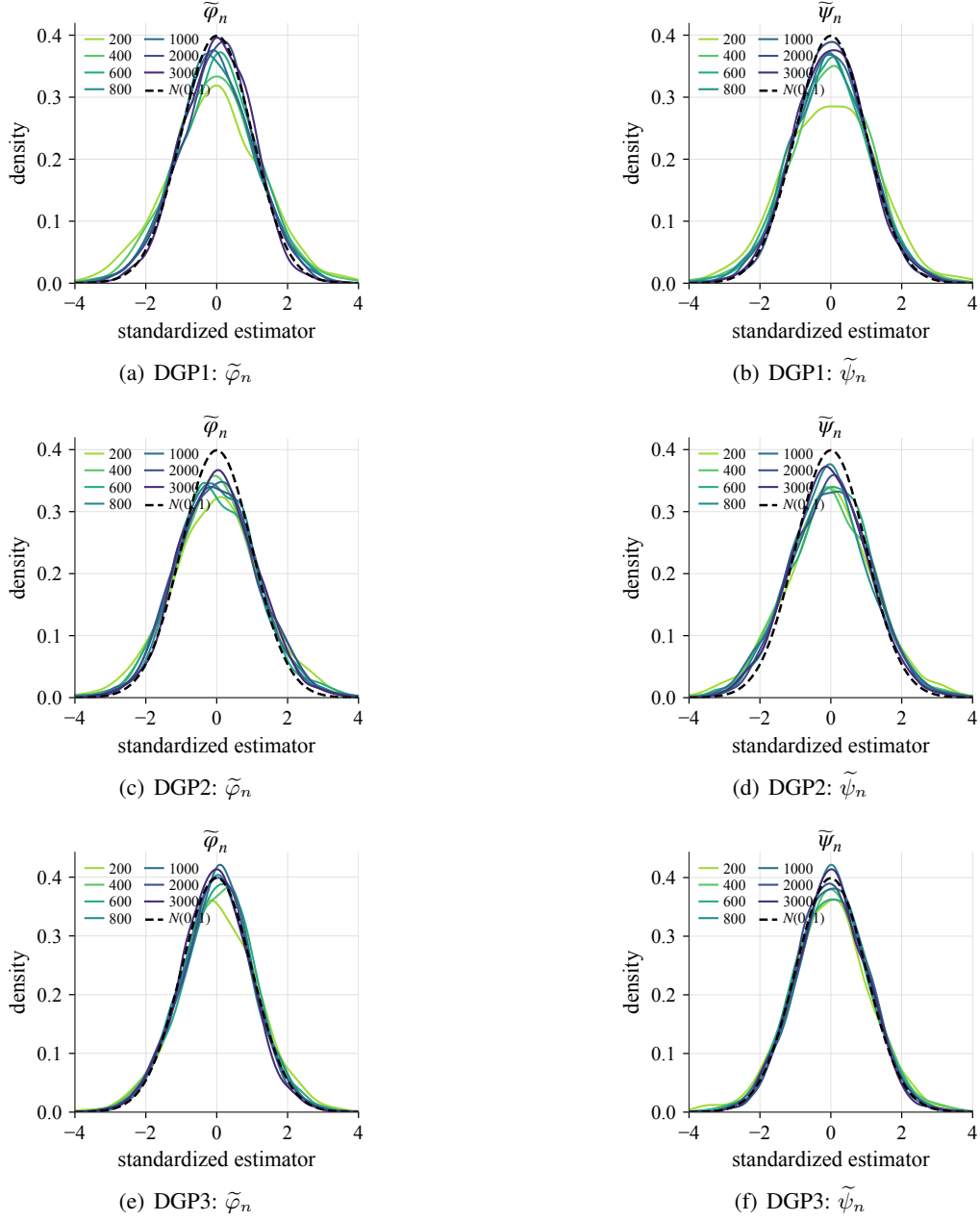


Figure A.3: SAMPLING DENSITIES OF THE MARDL-EC ESTIMATORS UNDER DGPs 1–3. Notes: (i) seven sample sizes from $n = 200$ to $n = 3,000$, with $R = 1,500$ iterations at each sample size. (ii) Each estimator is standardized by its asymptotic standard error, and the dashed curve is the standard normal density. (iii) DGP 1: $u_t \sim_{\text{iid}} N(0, 0.25)$; DGP 2: $u_t \sim_{\text{iid}} \text{Beta}(2, 6) - 1/6$; DGP 3: $u_t \sim_{\text{iid}} \Gamma(3, 1) - 2$.

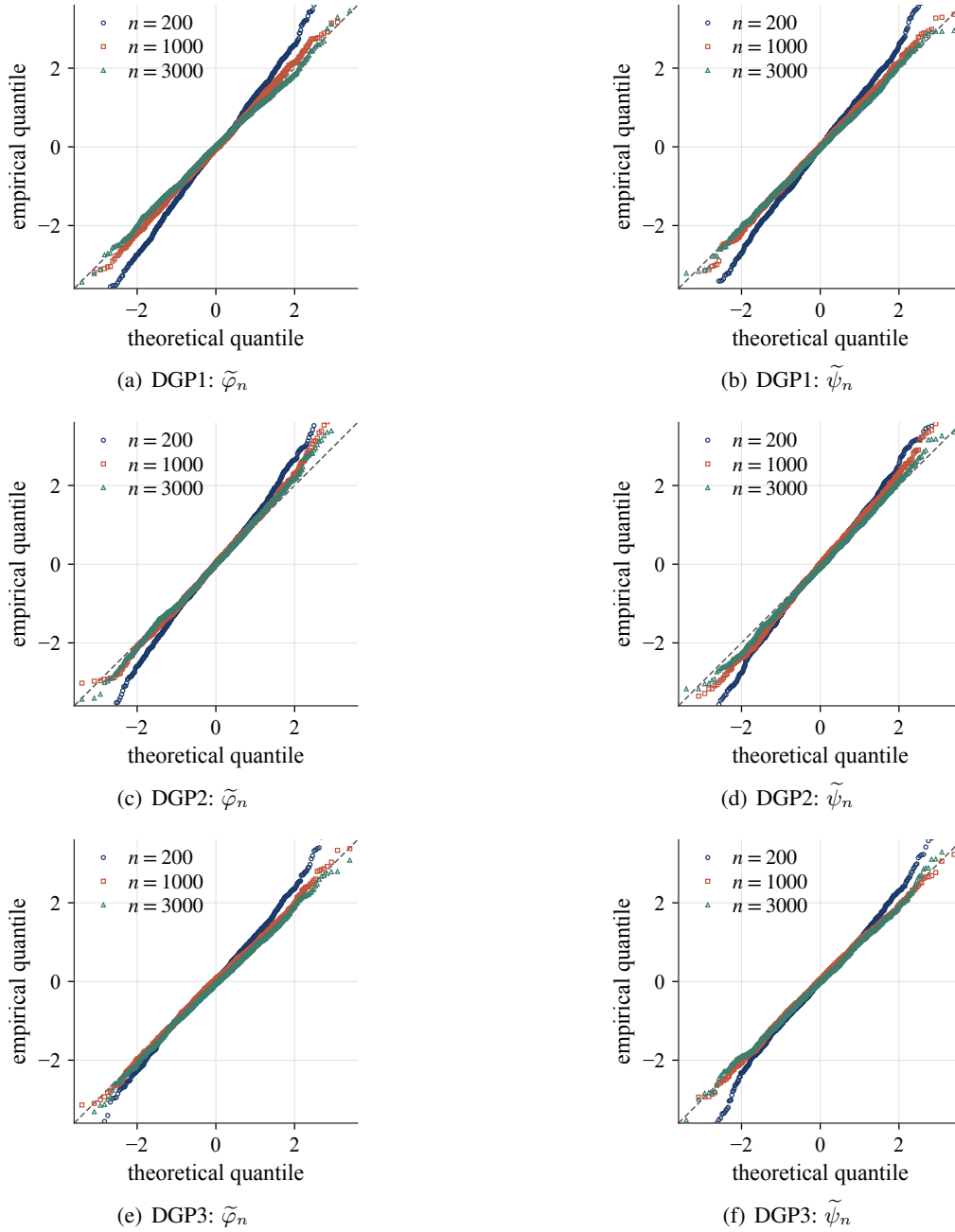


Figure A.4: QQ PLOTS OF THE STANDARDIZED MARDL-EC ESTIMATORS UNDER DGPs 1–3. Notes: (i) $R = 1,500$ iterations at each sample size. (ii) Each estimator is centered at its probability limit and divided by its estimated asymptotic standard error. (iii) The horizontal axis gives standard-normal quantiles and the vertical axis the quantiles of the simulated standardized estimator; the dashed line is the 45-degree line. (iv) The plots approach the 45-degree line as the sample size increases, with the remaining discrepancies concentrated mainly in the tails. (v) DGP 1: $u_t \sim_{\text{iid}} N(0, 0.25)$; DGP 2: $u_t \sim_{\text{iid}} \text{Beta}(2, 6) - 1/6$; DGP 3: $u_t \sim_{\text{iid}} \Gamma(3, 1) - 2$.